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MSc. Data Science  
7150CEM – Data Science Project  
Classification and identification of cervical cancer using CNN methods

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Submitted in partial fulfilment of the requirements for the Degree of Master of Science in  
Data Science  
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## Abstract

Cervical cancer emerges as a significant worldwide health issue, exerting an adverse effect on the overall death rate among women. Early and accurate detection is crucial for achieving successful therapeutic interventions and improving patient outcomes. Liquid-based cytology (LBC) is a potentially advantageous technique for the collection and preparation of cervical samples, as it has been shown to provide superior image resolution compared to traditional approaches. The primary aim of this study is to develop an automated diagnostic system for cervical cancer utilizing LBC Pap smear images, capable of classifying multiple classes.

The suggested methodology uses deep learning and machine learning algorithms to analyse and categorise cervical cell abnormalities. First, a sizable collection (963 images) of LBC Pap smear pictures will be assembled for training and assessment purposes and annotated by experienced cytologists. We just download the dataset that is already approved by cytologists. Here the benchmark dataset is downloaded from Mendeley website.

The next step is to design and refine a convolutional neural network (CNN) architecture for the multi-class classification of cervical cell types, including Negative for Intraepithelial malignancy, Squamous cell carcinoma, high squamous intra-epithelial lesion and low squamous intra-epithelial lesion. The pre-trained model's information is very useful in various medical image processing tasks, will be used to train the CNN (VGG-16, VGG-19, ResNet-50 and ResNet-101) and sequential classifier via transfer learning.

The suggested system's reliability and generalizability will be ensured by thorough training and testing on a Herlev dataset using the CNN model. Precision, recall, accuracy, and F1-score will be used to assess the model's performance, with a focus on achieving high sensitivity to reduce false negatives, which is crucial for cancer detection. The research demonstrates the use of an ensemble classifier that is comprised of for all the deep learning models in order to produce a multi-class classification with a high degree of accuracy.

Here, we found that among the four models VGG-16 and VGG-19 both are the best model for the image classification which has the highest accuracy of 94% whereas the next best model is ResNET 101 with accuracy of 80%.

The goal of this research is to improve cervical cancer diagnosis by utilising the potential of liquid-based cytology Pap smear images and the best deep learning algorithms. This research aims to increase women's healthcare outcomes and save lives through early diagnosis and intervention by offering accurate and effective multi-class classification.

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# 1 Introduction

Cervical cavity cancer continues to affect millions of women and has a big impact on mortality as well as morbidity rates across the world. Certain strains of the human papillomavirus (HPV) have the potential to induce unusual alterations in cervical cells, that ultimately result in the development of cervical cancer. Early identification of the cervical cancer and the disorders that precede is vital for a successful treatment plan and better patient outcomes.

The use of standard screening methods, such as typical Pap smears, has helped to detect abnormal cervical cells. Liquid-based cytology (LBC) has gained favour as a more sophisticated and dependable method for gathering and preparing samples. It offers higher resolution images and minimises specimen artifacts.

This project attempts to develop an automated multi-class cervical cancer diagnostic method using LBC Pap smear images. The efficacy and accuracy of screening for cervical cancer could be considerably improved with the help of deep learning and other technologies like artificial intelligence (AI) and Federated Learning.

## 1.1 Background to the Project

Traditional Pap smears have been the main form of cervical cancer screening, but they have some drawbacks such as a low degree of sensitivity and the possibility for sample deviations, which can lead to false-negative results. These limitations prompted researchers to look for alternative screening techniques, which led to the development of liquid-based cytology (LBC) as a more advanced procedure. LBC overcomes many of the issues with conventional Pap smears by keeping a more representative sample of cervical cells and decreasing unclear factors like blood and inflammation. The precision of cytological analysis is much increased and sample appropriateness is greatly improved by this method. In spite of this, evaluating LBC pap smear images is still a time-consuming operation that mostly relies on the judgement of expert cytologists and is vulnerable to inconsistency between different observers.

The proposed research intends to address these concerns and enhance the efficacy and accuracy of cervical cancer screening by developing an automated multi-class detection system using deep learning techniques and CNN (Convolutional neural network). By the automated processing of LBC Pap smear images, the research tries to speed the identification of pre-cancerous lesions and early-stage cervical cancer, which results in timely intervention and improved patient outcomes. The suggested approach may also be highly effective in low resource settings at which access to qualified cytologists may be constrained, giving medical professionals a potential tool for accurate diagnosis, reducing the prevalence of cervical cancer, and ultimately saving lives.

## 1.2 Project Objectives

The main goal of this study is to develop a reliable and scalable approach that categorise various cervical cell abnormalities using LBC Pap smear pictures. The proposed approach will cover a variety of diagnostic subtypes, including invasive cervical cancer, normal, squamous cells carcinoma, low-grade and high-grade squamous intraepithelial lesions, and negative intraepithelial malignancy. In order to make an early discovery, it is necessary to combine correct patient care with specialized treatment strategies in an appropriate manner.

For training and assessment purposes, a sizable and thoroughly annotated dataset of LBC Pap smear pictures are utilized in order to meet this objective. The cell images have been properly annotated by the experienced cytologists to provide accurate data. Performing exploratory data analysis on the dataset to see if there are any imperfections, noise, or if it is executing effectively. Effective deep learning and machine learning model training will depend on the utilisation of a correctly selected dataset.

Convolutional neural networks (CNNs), a class of deep learning algorithms renowned for their outstanding performance in image processing applications, are at the core of the proposed system. In order to analyse LBC Pap smear images and efficiently extract pertinent features and patterns indicating various cervical cell types, the various CNN architectures are used and optimised as per the data which is the aim of our research.

The objectives of our project are as below:

1. Downloading the dataset and performing exploratory data analysis on it.
2. Understanding the background on the Cervical cancer, its causes and identification.
3. Implement simplified CNN models for diagnosis of cervical pre-cancerous as well as cancer lesions.
4. Investigate the different structures of cancer cells and then sequentially categorise them to their corresponding classes they are associated with using ML models.
5. Investigate misclassified cases further, using boosting methods.

Through the use of liquid-based cytology and deep learning, this study has the potential to enhance cervical cancer screening and detection. An automated multi-class diagnosis system that is successfully implemented leads to the earlier detection, quicker interventions, and higher survival rates. In the end, our research aims to support efforts to eradicate cervical cancer and improve healthcare for women around the world.

## 1.3 Overview of This Report

The present report was created in the context of detecting cervical cancer cells. The current study includes a thorough analysis and advancement of all important domains like machine learning and neural networks that were investigated and cultivated throughout the project research and development efforts. The primary focus of this study includes the fundamental concepts of operation, detailed evaluations of existing literature, actual implementations, methodological approaches, and potential routes for future developments.



The report begins with an explanation of the project's contextual framework, which includes the underlying reasons and justifications that led to the project delivery (chapter 1). Furthermore, the document describes the project's goals and objectives (in chapter-1.2), as well as the intended outcomes and the related actions required to attain them. The subsequent section (chapter 2) of the paper is literature review, which encompasses a range of studies, prior experiences associated with identical challenges, and the resultant outcomes derived from those interactions.

In chapter 3, after conducting an in-depth analysis of the relevant literature, the next part of this study explains about the detailed examination of the algorithms and methodologies utilized to address the research problem, including the specific type of neural network and it's layers that was employed, among the other relevant factors. Subsequently, a distinct section was incorporated to undertake an analysis that includes diverse factors, methodologies, and other pivotal components. The design part covers both the algorithm's architectural design and its functionality. The implementation section describes the complexity of execution and realization of each individual task. The coding part was fully composed and elaborates its underlying principles and fundamental principles followed by results section.

Any potential management issues that can arise while the project is being carried out are covered in the section on project management (Chapter-9). Finally, in chapter-11, student reflection has been included to emphasize what has been learnt and accomplished from the research. The conclusion's successes and future work are described here.

#### **1.4 Research design and outline principal methods.**

The goal is to use TensorFlow and keras to categorize images into their appropriate classes. Due to the impending challenges with running the image dataset on the local PC, I will consider using a GPU service for example Google to be used with the Colab IDE with V100 GPU.

Convolution neural network (CNN) models will be applied to an existing dataset of cervical cancer images, for example VGG (vgg16, vgg19), resnet (resnet 50, resnet 101).

## 2 Literature Review

Cervical cancer incidence and mortality rates have been demonstrated to decrease with screening programmes that incorporate cytology and HPV (human papillomavirus) testing)

Castle, et al., (2005).

The most popular screening tools for identifying precancerous and cancerous cervical lesions are the Pap smear and LBC (Liquid based cytology) Arbyn, et al., (2008).

The combination of molecular biomarkers with LBC images, in addition to image analysis methods, has demonstrated potential for multiclass cervical cancer diagnosis. Cervical lesions can be detected and tracked using molecular biomarkers, such as human papillomavirus (HPV) DNA testing and RNA expression monitoring Schiffman et al., (2011).

Castle et al., (2011) combining LBC pictures with molecular indicators can increase the accuracy of diagnosis overall, help with personalized treatment decisions, and improve risk identification and classification.

Particularly in areas with little resources, cervical cancer continues to be a major worldwide health concern (Kim, et al., 2013). With over 570,000 new cases and 311,000 reported deaths each year, it is the fourth most prevalent cancer in women worldwide (Bray et al., 2018). Since there is less access to screening and other preventative measures in underdeveloped nations, the burden of cervical cancer is disproportionately higher there. Since symptoms frequently appear in late stages, when there are few treatment options available, the high fatality rate linked with cervical cancer can be related to late-stage detection. Early diagnosis with efficient screening techniques is therefore essential for enhancing patient outcomes and lowering mortality rates (Kim, et al., 2013).

LBC pictures are sent to cytologists in the form of computerized images using telecytology for remote diagnosis and consultation Badaya, et al., (2016). This method is especially useful in underserved areas where access to knowledgeable cytologists is constrained. Tele-cytology provides real-time or store-and-forward transmission of LBC images using digital imaging and communication technology, facilitating rapid and precise diagnosis. Tele-cytology, particularly in areas where cytological knowledge is lacking, has the potential to eliminate healthcare inequities by granting access to expert perspectives and second opinions.

Higgins, et al., (2015) mainly concentrated on Virtual microscopy and explains that, acquiring, storing, and deciphering high-resolution whole-slide images (WSIs) of cytological samples are all part of digital pathology and hold immense promise in enhancing patient care and enhancing the pathology workforce. This report was presented in the form of an educational discourse. By streamlining the process of disease diagnosis and monitoring, digital microscopy has the capacity to significantly improve efficiency. LBC (Liquid Based Cytology) slides can now be digitally preserved as cytological image archives thanks to advancements in digital imaging technologies. Remote viewing, effective storage, and accessible sharing of LBC pictures are all made

possible by whole-slide imaging, which also makes remote consultation possible. Additionally, for increased diagnostic effectiveness and efficiency, digital pathology integrates AI algorithms and computer-assisted image analysis tools. The enhanced sensitivity and specificity that LBC offers over traditional Pap smears is one of its main advantages. LBC has demonstrated to reduce the percentage of poor-quality samples, enhance the detection of glandular lesions, and provide opportunities for additional molecular testing.

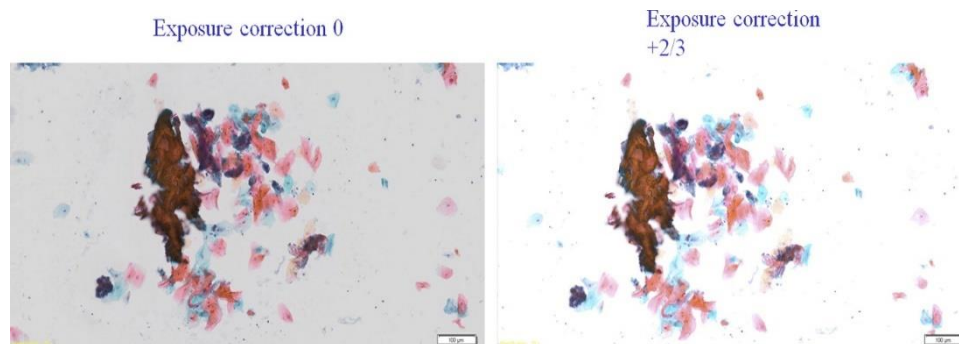


Fig1. Shows the value of gathering the more samples as possible during an exposure.

Instead of conducting manual examination of numerous glass slides, Higgins employed virtual microscopy as a means to expedite the advancement of novel procedures. This approach involves the utilization of automated methods, which rely on image preprocessing, feature extraction, and machine learning techniques to achieve dependable and unbiased diagnoses. By employing these techniques, the early detection and accurate classification of cervical lesions can be facilitated.

According to Hussain, et al.,(2020), when the patient to pathologist ratio is high, commercially accessible and modern research-based decision support systems are better equipped to understand the diagnosis of cervical dysplasia, carcinoma in situ (a cluster of strange cells that have not yet progressed beyond the site where they formed, but which have the potential to develop into cancer) , and confirmed carcinoma cases. The precise dysplasia stage classification derived from the Bethesda System is the sole foundation upon which treatment options for this diagnosis may be constructed. The Bethesda System categorization is a multiclass problem that is extremely important and significant because of its consistent and uniform distribution. When image interpretation is done manually, interobserver variability (The variation observed in multiple measurements conducted by a single observer) is introduced, which makes microscope viewing tiresome and time-consuming.

Hussain implemented a deep learning, Alex-net and CNN based computer-assisted screening architecture to data for classification which helps pathologists screen with accurate predictions at a faster rate. With the development of artificial intelligence in the healthcare industry, decision support system forecasts can address difficulties with observer biases. To find any squamous intraepithelial lesions (A confined appearance of atypical cellular growth that arises on the epithelial surfaces of specific organs, including but not limited to the cervix, vagina, vulva, anus, and oesophagus) or malignant development, a Pap test is utilised as a cervical cancer screening technique. The Pap test report can serve as an example of early detection of cervical dysplasia or squamous intraepithelial lesions, which can be treated quickly to stop it from progressing into invasive cancer.

Hussain addresses the issue of inaccurate predictions, eliminating the need for segmentation and manual feature extraction steps that are typically found in traditional conventional methods. Without relying on earlier pipelined techniques like segmentation algorithms followed by feature extraction and feature selection approaches, the current study offers entire slide image analysis. Through the use of deep learning, Alex-net and CNN, this study was able to automatically diagnose and predict the cervical cancer class from healthy or normal, dysplasia, and carcinoma samples. The experimental results show that the proposed ensemble classifier outperforms other alternative classifier models in terms of accuracy, precision, and recall in predicting the four target classes (NILM, LSIL, HSIL and SCC). Patients with low-grade and high-grade squamous intraepithelial lesions have a high risk of developing cervical squamous cell carcinoma or potentially invasive carcinoma if they are not promptly diagnosed. A pathologist can frequently check lesions thanks to the pap smear screening procedure.

Sigurdsson, et al., (2013), explain that Liquid-based cytology (LBC) has become a popular Pap smear substitute because it has a number of benefits. In LBC, the gathered cervical cells are suspended in a liquid medium and treated to produce a thin layer of cells for analysis. By minimizing concealing elements including blood, mucous, and inflammation, this approach enhances the sample's quality. Human papillomavirus (HPV) testing, which is a high-risk factor for the development of cervical cancer, is one of the additional tests that LBC permits. LBC images can also be archived for later analysis, improving quality control and retrospective research. Despite these benefits, it takes specialist expertise and can often be subjective to evaluate LBC Pap smear images. Delays in therapy and negative results might result from inadequate or missed diagnosis of precancerous or cancerous tumours. There is a need for impartial and trustworthy techniques to enhance the diagnostic precision and effectiveness of LBC Pap smear analysis. In conclusion, the findings of this study provide credence to the theory that Pap smears are more sensitive than LBC tests for recognizing HSIL and CIN2 lesions, regardless of the patient's age. LBC was found to reduce the number of insufficient smears, but it was found to increase the number of poor-quality cytology cases in patients who are below 40 years, while reducing the number of total negative smear cases in patients over 40 years.

Pradeep, et al., (2019) explains that image processing plays a crucial role in the realm of medicine, where its extensive utilization aims to enhance the quick detection and subsequent management of various conditions. Deep Neural Networks (DNN) have demonstrated remarkable performance in tasks involving classification and segmentation. In consideration of this, the researcher introduced a novel approach for reducing image size known as the Deep Wavelet Autoencoder (DWA). This method effectively merges the image breakdown capability of the wavelet transform with the core reduction of auto-encoder features. The utilization of the two approaches has resulted in a significant decrease in the necessary features for future classification tasks involving deep neural networks (DNN). A comparative analysis was conducted for the performance evaluation of the DWA-DNN classifier in relation to other established classifiers, including autoencoder-DNN and DNN which demonstrates superior performance in image processing and classification.

In the research they conducted, Chandran et al. (2021) introduced two convolutional neural network (CNN) architectures, namely VGG19 (TL) model and CYENET, for the purpose of identifying cervical cancer through the analysis of colposcopy images. The VGG19 model was employed as a transfer learning technique in the research on convolutional neural network (CNN) architecture. The experimental results showcased

the exceptional performance of the proposed CYENET. The CYENET model exhibited a significantly higher classification accuracy, overtaking the VGG19 (TL) model by an impressive margin of 19%.

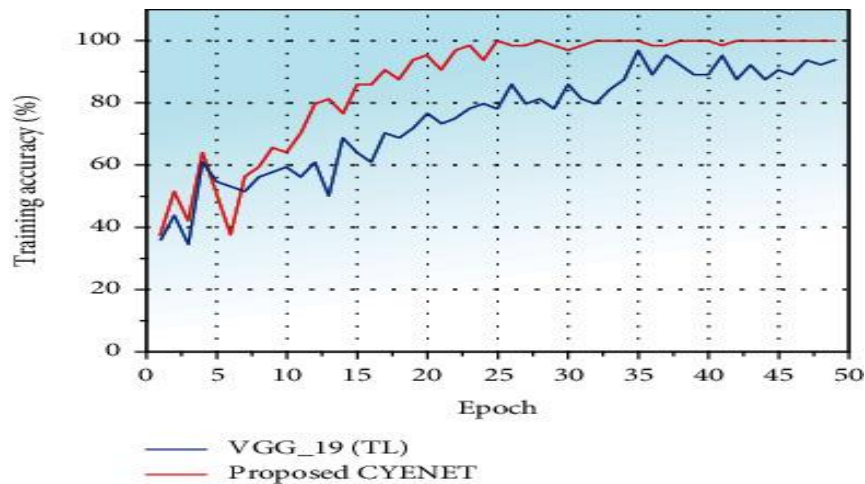


Fig 2. Accuracy for VGG-19 and CYENET.

Amjad, et al., (2023) highlights the promise of automated image analysis approaches for enhancing diagnosis accuracy and minimizing healthcare inequities as it analyses the advantages, limitations, and technological breakthroughs in this area. Different cytologists' manual interpretations of cytological images can differ and are subject to subjectivity. Due to this subjectivity, there is a chance for false positives and negatives, which could result in overlooked or incorrectly diagnosed cases. Automated image analysis methods seek to improve consistency and objectivity of diagnoses while reducing inter-observer variability (Tu, et al., 2017).

Additionally, the large number of samples that screening programmers receive makes manual interpretation difficult, which frequently results in time restrictions and the possibility of accuracy compromise. The procedure can be sped up using automated picture analysis, which is CNN (Convolutional Neural Network) based computer aided diagnosis (CADx) method, enabling effective screening and triage of aberrant situations. LBC has a number of benefits over a traditional Pap smear. LBC has been demonstrated to improve the detection of glandular lesions, boost sensitivity and specificity, decrease the number of unsatisfactory samples, and offer the option for additional genetic testing (Deshou, et al., 2009). Additionally, LBC supports computer-aided analysis, enabling automated image processing and analysis methods to boost diagnostic precision.

Hou, et., al (2022) Using LBC pictures to diagnose cervical cancer may be changed by a number of new trends. These trends incorporate whole-slide imaging and digital pathology, which allow for remote viewing, effective storage, and sharing of LBC images. Tele-cytology enables remote diagnosis and consultation, which is advantageous in underprivileged areas in particular. Multi-class diagnosis is more accurate and effective when artificial intelligence (AI) and machine learning methods like deep learning and transfer learning are combined. Additionally, efforts are being undertaken to guarantee the interpretability and explicability of AI models in order to increase doctors' and cytologists' acceptance and faith in them.

Arora et al., (2021) describe image pre-processing is one of the crucial steps in using machine learning to diagnose cervical cancer. Image enhancement and noise reduction techniques are employed to enhance the quality of Pap smear images, thereby

facilitating the identification of abnormal cells and structures. To isolate cell nuclei and cytoplasm and enable precise feature extraction, segmentation techniques are applied. To ensure the accuracy of the analysis, artefact removal techniques are used to remove undesired aspects from the photos. Additionally, quality assessment techniques aid in locating photos of poor quality that can compromise the accuracy of the diagnosis.

Krithiga, & Geetha et.al., (2021), shows that obtaining the data from Pap smear images, feature extraction and selection are essential. Cell nuclei's shape, size, and texture, among other morphological and textural characteristics, offer important classification-related data. In order to distinguish between distinct classes, statistical parameters like mean, standard deviation, and entropy capture the distribution properties of cell nuclei. Convolutional neural networks (CNN)-based deep learning-based feature extraction has drawn a lot of interest since it can automatically learn hierarchical characteristics from photos. Within the scope of this investigation, an independent Bayesian classification approach is utilized as a method for successfully distinguishing and eliminating the overlapping nuclei from the environment of cancer cells. The diagnostic accuracy is increased (from 77.8% to 83.3%) by this deep learning models' efficient extraction of distinguishing information from Pap smear images.

Wong, et al., (2023), states that due to insufficient screening, one woman worldwide dies of cervical cancer every two minutes. The liquid-based cytology (LBC) technique was created to test 100% of the sample and increase accuracy because the Papanicolaou technique (which is also called as pap smear technique) only examines 20% of a sample, which results in a large number of incorrect results. Here, totally 2,676 images are used. In this study, the researchers have successfully used the machine learning like SVM, random forest and logistic regression to train the model. This system has the capability to accurately detect the lesion level of cervical cancer in liquid-based Pap smears, utilizing the widely recognized Bethesda classification of cancer. Two models, namely ResNet50V2 and ResNet101V2 were developed utilizing the principles of deep learning and transfer learning methodologies (Tensor flow and Keras tools) and optimizing each model by determining Early Stopping hyperparameters (assisted in the reduction of execution time and resource consumption, while effectively mitigating the issue of overfitting in the models). The evaluation revealed that the ResNet50V2 model demonstrated superior performance, particularly in the classification of HSIL (High grade squamous intraepithelial lesion) and SCC(Squamous cell carcinoma) type images. It achieved a precision of 0.98 and an accuracy of 0.97. The construction of the system utilizing the ResNet50V2 model has been completed, and a comprehensive validation of its performance has been conducted.

In the study conducted by Vidya, et al., (2016), a comprehensive exploration was undertaken to investigate various machine learning techniques for the purpose of cervical cancer prediction. The employed methodologies encompassed classification and regression tree, random forest tree, and random forest tree with k-means learning on 917 images, those are extracted from UCI. The study achieved a remarkable maximum accuracy of 96.7%. In particular, a full collection of 61 features was carefully prepared for this purpose among these the useful features are size, shape, colour, texture and depth and number of trees.

Using the BreakHis dataset with transfer learning, Murtaza et al., (2019) suggested a precise model for classification purposes. The architecture employed in this study is based on AlexNet, a widely recognized convolutional neural network. To optimize the model for the specific task at hand, the final layer is fine-tuned to enable accurate

classification of binary data. Six distinct machine learning classifiers (softmax, k-nearest neighbour, Naïve Bayes, Linear discriminant analysis, Decision tree, Support vector machine) are subsequently employed to utilize the acquired features for the purpose of performing classification into two distinct classes. Additionally, the proposed algorithm aims to reduce inaccurate categorization. The experiment yielded an accuracy rate of 81.25% instead of using the above-mentioned methods, CNN architecture and ADA booster can be used in-order to gain more accuracy.

As per Marklew, et al., (2022) gauge the effectiveness and generalizability of machine learning algorithms for cervical cancer diagnosis must be evaluated. Accuracy, sensitivity, specificity, and area under the curve (AUC) are a few of the evaluation metrics used to assess the efficacy of classification algorithms. Cross-validation methods and independent validation on external datasets are widely employed to assess the efficacy of the developed models. Comparative studies have also been conducted to assess the effectiveness of various algorithms and identify the most accurate and reliable techniques. Area under the receiver operating characteristic curve (AUC-ROC) and other metrics are used to assess the effectiveness of automated image analysis systems for cervical cancer diagnosis. And these studies examining the efficacy and generalizability of machine learning algorithms for cervical cancer diagnosis have produced encouraging results. Comparative studies (when compared with cytological pap smear and HPV screening methods) have shown that automated analysis approaches perform better in terms of diagnostic accuracy, sensitivity, and specificity than traditional Pap smear. Additionally, comparative studies have been conducted to evaluate the efficacy of various automated analytic methods and highlight their benefits and drawbacks.

Marklew, et al.,(2022) suggests that, even while advancements in automatic image processing techniques are encouraging, a number of problems still need to be fixed. To evaluate the effectiveness, generalizability, and reliability of these approaches, robust research involving sizable and varied patient groups are required. Careful validation, regulatory compliance, and consideration of ethical and legal ramifications are all necessary for integrating automated technologies into ordinary clinical practice. For the successful translation and application of these strategies, cooperation between researchers, physicians, regulatory agencies, and policymakers is essential. Although improvements in image processing methods for cervical cancer diagnosis are encouraging, their clinical validation and incorporation into standard practice.

Rajaram & Gupta, et al., (2021) describes many challenges in the interpretation of manual cytological image, as manual cytological image interpretation is a subjective process that varies among cytologists based on the outputs they get like accuracy, classification, speed, threshold. Automated image analysis methods seek to improve consistency and objectivity of diagnoses while reducing inter-observer variability. Additionally, the large number of samples that screening programmers receive makes manual interpretation difficult, which frequently results in time restrictions and the possibility of accuracy compromise. The procedure can be sped up using automated picture analysis, enabling effective screening and triage of aberrant situations.

Wang, et al., (2019), states that due to new developments and upcoming directions Convolutional neural networks (CNNs) and deep learning, have made recent strides that have showed promise in the diagnosis of cervical cancer. Convolutional neural networks (CNNs) have shown remarkable proficiency in accurately classifying and detecting abnormalities through the automatic acquisition of hierarchical features from



cytological images. The accuracy of cervical cancer detection can be improved by combining molecular indicators with liquid-based cytology, such as HPV DNA testing and RNA expression profiling. Particularly in underprivileged areas, tele cytology, which allows for remote diagnosis and consultation, can provide access to qualified cytologists. In cervical cancer screening and diagnosis, patient-centred methods and addressing healthcare disparities should be taken into account. Hence automated cytology of liquid samples Pap smear images has a significant deal of potential to increase the accuracy of cervical cancer diagnosis. These methods can help cytologists make diagnoses that are more consistent and reliable by lowering subjectivity and inter-observer variability. To address issues, validate performance, and assure the effective incorporation of automated systems into standard clinical practice, more research and development is needed.

Karri, et al., (2022) claim that, using cell image analysis to categorize cells as normal or abnormal, the liquid-based cytology (LBC)/Pap smear method of enhanced cervical screening is a highly successful it detects the precancerous cell detection tool because of its high accuracy and speed. This study explains the limitations by presenting a novel paradigm for the correct classification of cervical cells. Segmentation, localization of the nucleus, and classification are the three phases of the proposed methodology. In the segmentation stage, researcher developed a hybrid system that combines two binary image patches acquired using a 19-layered convolutional neural network (ConvNet) model with enhanced deep high dimensional dissimilarity translation (HDDT) based conspicuous segmentation. The recommended technique is known as optimum semantic similarity selective search (OSS-SS), and it retrieves the localised RGB data from binary patched images. To reduce redundancy in nuclei classification, ResNet-50 ConvNet was down-sampled using t-SNE (t-distribution stochastic neighbour embedding) feature reduction. This methodology effectively eliminates a significant portion of healthy cells, specifically the middle and superficial epithelium, from the sample. This process serves to minimize the amount of effort required for focused reading and also reduces the likelihood of screening errors. Limitations in this research are using small dataset and there is a class imbalance with few patches.

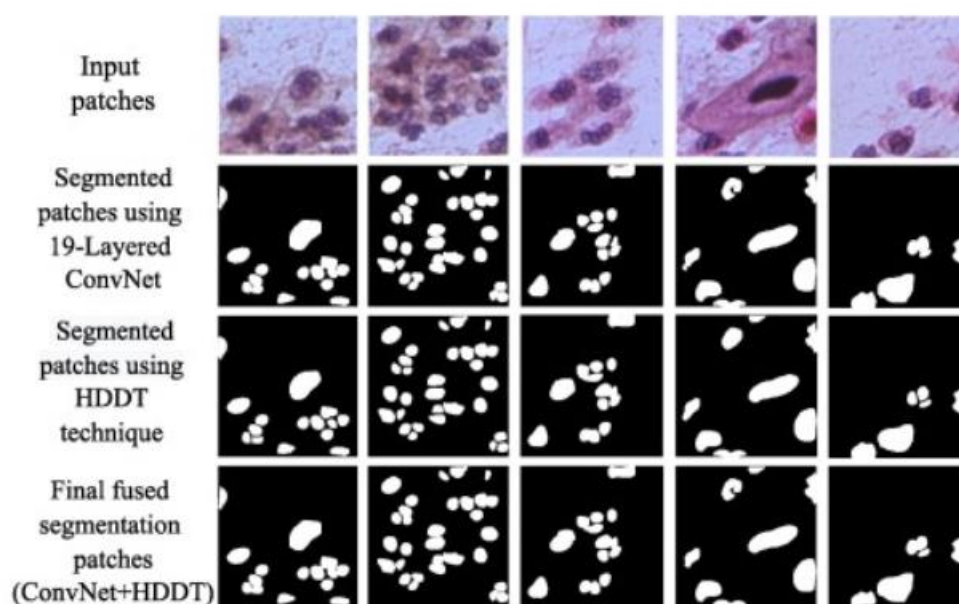


Fig 3. Visualization results post applying CNN.



Also, Xue, et al., (2020) says, machine learning and computer vision are two examples of artificial intelligence tools that have the potential to increase the accuracy of cervical cancer diagnosis. To find aberrant cells and patterns linked to cervical cancer, Xue have created a practical, precise AI-guided digital colposcopy (If any abnormal cells are discovered in HPV during cervical screening, a colposcopy will be performed to examine the cervix) may improve cervical biopsy diagnostics, increase cervical cancer screening in LMICs, and speed global cervical cancer elimination. These methods make use of feature extraction, classification, and picture segmentation algorithms that have been trained on big datasets (more than thousand images). It is quite likely that additional study and advancement in this field will improve cervical cancer screening and ultimately result in lifesaving outcomes. Machine learning can help to evaluate LBC images, discover trends, and group the images into several diagnostic categories. Support vector machines (SVMs), artificial neural networks (ANNs), and random forests are a few of the machine learning methods that are widely applied in this area. Challenges including the small dataset size standardized picture acquisition, and integration into clinical workflows still exist, though. Despite these difficulties, it is clear that computer vision and machine learning have the potential to improve cervical cancer diagnosis. Ongoing research attempts to get over these obstacles and pave the road for their practical application in routine screening programmers. By recognizing a number of cellular abnormalities, these algorithms enable reliable and early detection of cervical cancer. The automatic analysis of liquid-based cytology images has been identified as a viable strategy to improve cervical cancer diagnosis precision. To enhance image quality and retrieve pertinent information, image preprocessing techniques like noise reduction, contrast improvement, and image segmentation are used. To identify cellular abnormalities related to cervical cancer, many features are retrieved from cytological pictures, such as morphological, textural, and nuclear properties. Cervical cytological image classification has been applied to machine learning and pattern recognition algorithms like support vector machines (SVM), random forests, and deep learning techniques. These algorithms categories fresh, unknown samples using labelled training data.

Fig. 1

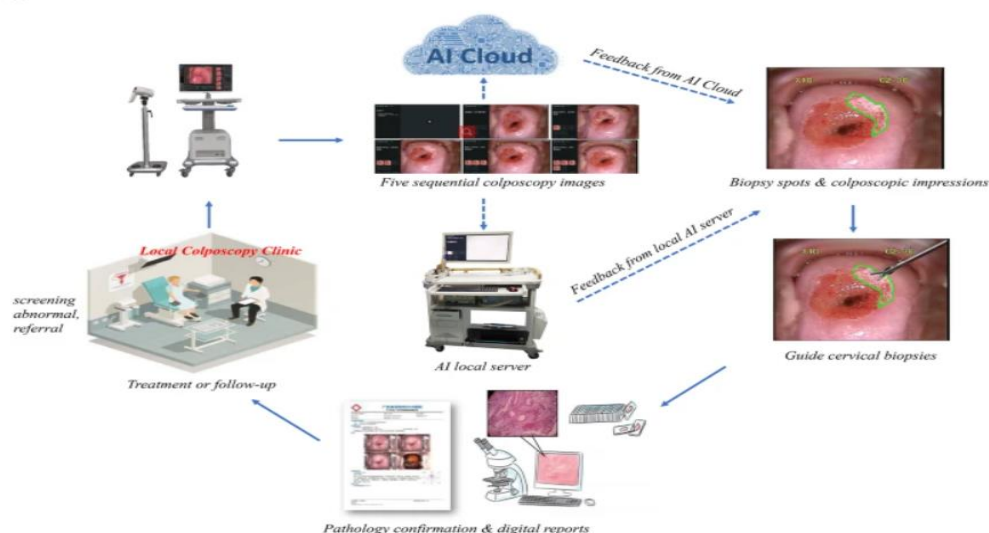


Fig 4. Diagnosis workflow by author (Xue)

In a recent publication by Huang et al. (2022), a detailed review of cervical colposcopy imaging features was conducted by using a dataset with 10,012 images which is gathered from 1,081 HPV patients. The study focused on five distinct categories, namely cancer, polyps, intraepithelial lesions, chronic cervicitis, and free hyperplastic

squamous epithelial tissue. These items were extracted for utilization in their convolutional neural network model designed for deep learning networks. The findings, nonetheless, revealed a significant decrease in accuracy (42.16%) due to the computer's inability to accurately distinguish intraepithelial lesions, free hyperplastic squamous epithelial tissue, and chronic cervicitis, all of which exhibit a considerable degree of resemblance. The researchers carefully selected two significant feature images, namely chronic cervicitis, and cervical cancer, which were then inputted into a deep learning network with the aim of optimizing the model. The results demonstrated a remarkable accuracy of 95.19%, indicating a high level of precision and resilience (enhancing the adaptability of the model, by incorporating two additional feature images into the network layers). utilizing the colposcopy images, this particular procedure serves as a valuable tool in discerning the presence of chronic cervicitis or cervical cancer within the patient.

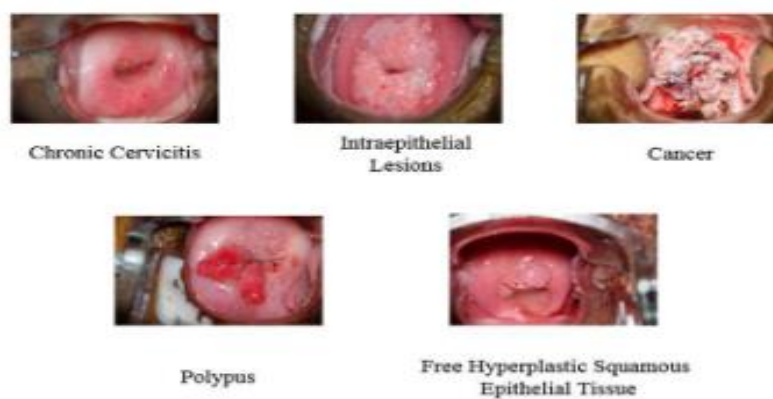


Fig 5. Sample images for each category.

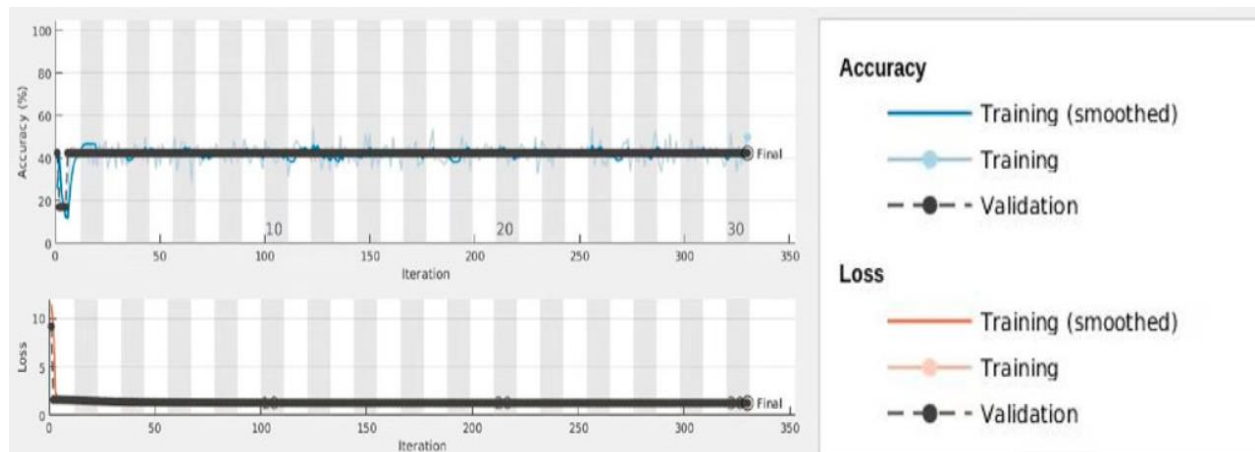


Fig 6. Accuracy and loss for training and validation data for all classes.

Overall, the literature review performs a number of tasks, such as highlighting the significance of cervical cancer as a global health concern, outlining the difficulties with traditional cytology techniques, talking about the benefits of LBC and automated image analysis, and investigating the potential of Machine learning techniques in improving cervical cancer diagnosis. In order to address healthcare inequities, the assessment also recognises the value of remote consultation and decision-support systems, particularly in regions with insufficient access to specialised cytologists.

### 3 Methodology

#### 3.1 Data set

In this study, we used LBC along with machine learning approach that has seen significant use in the medical domain. Our particular interest is to detect and perform classification types of cervical cancer cells (NIM, HSIL, LSIL, SCC). Additional research initiatives include Convolutional Neural Networks (CNNs), TensorFlow, and Transfer Learning, which are being undertaken with the deep learning methodology VGG16, VGG19, Resnet 50, and Resnet 101. Previous tasks included data collection, research, and analysis. It has facilitated the acquisition of substantial knowledge on how to effectively address this particular challenge.

The academic institution has granted ethics report clearance for the current effort. As a result, the project's perspectives, methodology, and initiatives indicate adherence to the university's policies.

A freely accessible benchmark dataset serves as a foundation for the advancement of innovative algorithms and the evaluation of outcomes. However, it is equally crucial to incorporate hospital-based data obtained from real-world healthcare environments which helps in finding the automated methods for medical research. This data is particularly valuable for disease diagnosis, categorization of predicted diseases, and class classification according to the protocols. By incorporating such data, the developed algorithm can strive for maximum accuracy. This dataset helps for fulfilling the image classification and segmentation tasks.

To compare the effectiveness of liquid-based cytology (LBC) with traditional Pap smears in the multi-class diagnosis of cervical cancer, this study used a retrospective observational design.

The dataset comprises of total of 963 images, which have been categorized into distinct groups based on the presence of low- and high-grade squamous cells, negative intraepithelial malignancy, as well as squamous cell carcinoma among them cells, they are further classified into cancerous and non-cancerous cells. The process of obtaining the pap smear images was carried out using a microscope at a magnification level of 40x. The classification of these images within the dataset has been accomplished through the utilization of the Bethesda system which describes the outcome of a pap smear test. The current study uses a 460 individual's samples. Each patient sample is tested using a microscope to find abnormal cells or changes in cell growth, which is a time-consuming process.

As a result, the introduction of automated diagnosis has the potential to improve the effectiveness of illness screening, allowing for the rapid beginning of timely treatment and medication. Hence, the adoption of this methodology has the potential to significantly mitigate the occurrence of delayed diagnoses in cases related to cervical cancer. Examples from Dataset Images are shown in fig 6, fig 7, fig 8, fig 9:

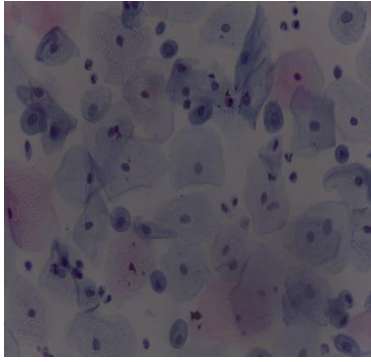


Fig 7. LSIL (Low squamous intra-epithelial Lesion)

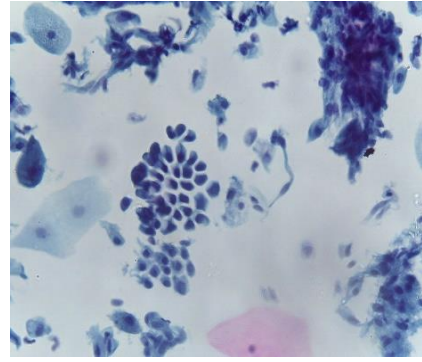


Fig 8. HSIL (High squamous intra-epithelial Lesion)

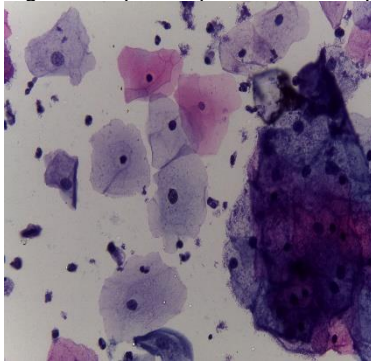


Fig 9. NIM (Negative Intraepithelial Malignancy)

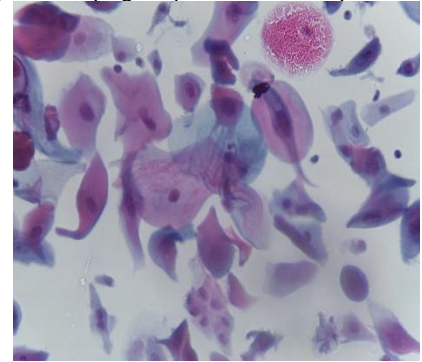


Fig 10. SCC (Squamous cell Carcinoma)

| Class                                           | Count of Images |
|-------------------------------------------------|-----------------|
| 1. Negative Intraepithelial Malignancy (NIM)    | 613             |
| 2. High Squamous intra-epithelial lesion (HSIL) | 163             |
| 3. Low squamous intra-epithelial Lesion (LSIL)  | 113             |
| 4. Squamous cell Carcinoma (SCC)                | 74              |
| <b>Total</b>                                    | <b>963</b>      |

Table1. Total number of images in each class.

### 3.2 Expert Commentary

To offer precise ground truth data, knowledgeable cytologists (a person who works in a lab on cell analysis) have thoroughly annotated each LBC Pap smear image. In order to guarantee the validity of the training and evaluation process, their competence is essential. Each image will have comments from skilled cytologists identifying the precise type of cervical cell that is present.

### 3.3 Data Preprocessing

Resizing (to enable effective processing by the CNN model, the collected images will be downsized to a standard resolution, such as 224x224 pixels), normalisation (Image normalization will be performed to scale pixel values within a specific range, e.g., [0, 1], to ensure consistent data representation), and noise reduction are some of the preprocessing stages that will be applied to the collected photos. The goal of preprocessing is to standardise the images and eliminate any distortions or artefacts that can impair the performance of the model.

The methodology chapter is summarized in this section, emphasizing the crucial actions required to accomplish the study's goals. It also shows how pertinent the approach is to be answering the study questions and how it can effect on how cervical cancer is diagnosed. The measures taken in the research study on employing liquid-based cytology Pap smear pictures for multi-class diagnosis of cervical cancer are described in this methodology chapter. To achieve accurate and dependable findings, each step was meticulously planned out and carried out. The work sought to contribute to the creation of an accurate and effective classification model for cervical cancer diagnosis by using appropriate dataset gathering and preprocessing, feature extraction, model selection, training and evaluation, and performance assessment procedures.

### **3.4 Feature Extraction by us**

On the pre-processed data, feature extraction will be carried out with the presumption that the selected computer vision and machine learning approaches demand particular characteristics for classification. Different image processing approaches will be used, presuming that the chosen elements, such as form, texture, colour, and intensity, are suitable for capturing the pertinent patterns and characteristics in the LBC Pap smear photos. Through the use of these techniques, the desired features from the photos will be extracted and used as inputs for the classification machine learning algorithms.

The LBC Pap smear pictures are then evaluated and classified independently by a group of skilled cytopathologists in accordance with the predefined classes. Each specialist carefully reviews the photos, spots any aberrant cells or characteristics, and determines the correct cervical cancer stage.

A computer-aided diagnostic (CAD) system is used by our team to further analyses the LBC Pap smear pictures. The CAD system automatically extracts pertinent information from the photos using cutting-edge image processing methods and machine learning algorithms. The cervical cells, nuclei, and surrounding structures are represented by a variety of morphological and textural traits. The CAD system is trained using LBC Pap smear images and professional diagnosis. Feeding photographs to the system during training teaches it to recognize and classify images using extracted features.

After training, the CAD system is tested with a separate set of LBC Pap smear photos. The trained system automatically analyses, extracts, and diagnoses these test photos. As per the accuracy parameter the system's ability will reflect effectively and classify LBC Pap smear pictures. Researchers carefully investigate false positive and false negative samples to find CAD system patterns and performance restrictions. This project improves the cervical cancer classification method, making it more accurate and useful.

### **3.5 Training and Testing data**

Machine learning models will be trained on the prepared data under the presumption that the extracted characteristics are reflective of the underlying patterns in the LBC Pap smear images.

In order to simplify data pre-processing, the dataset has been partitioned into two distinct subsets, namely the training set and the testing set. In the context of the training

process, it is noteworthy that a majority of the available data, specifically chooses 70-80% of data for training and 20-30% of data for validation but here we use 90% (in-order to maintain the class balance) for the purpose of training the model. Conversely, a smaller proportion, amounting to 10% of the data, was reserved for the purpose of testing the model's performance.

```
#splitting training and test data by preserving the class distribution:
train_data, val_data = train_test_split(data, test_size=0.1, random_state=42, stratify=data['class'])
print(train_data['class'].value_counts(dropna=False))
print(val_data['class'].value_counts(dropna=False))
```

```
2    550
0    147
1    102
3     66
Name: class, dtype: int64
2     62
0     16
1     11
3      8
Name: class, dtype: int64
```

Fig 11. Train and test data

### 3.6 Models

The selection of a suitable model should meet the aims and objectives that are stated in chapter 1.2 which was a critical component of this research. Based on the outcomes of the examined studies from chapter 2, it is clear that the Convolutional Neural Network (CNN) algorithm remains the dominant and highly proficient model used in image categorization tasks.

TensorFlow is a free software library for machine learning and artificial intelligence. The tool can be used for a variety of tasks, with specific skill sets for deep neural network training and inference. TensorFlow, a free and open-source software library, is a complete toolkit for a wide range of machine learning tasks. Keras, on the other hand, is a specialized neural network library.

TensorFlow provides a comprehensive range of APIs, encompassing both high-level and low-level functionalities. In contrast, Keras exclusively offers high-level APIs. Keras, being implemented in the Python programming language, provides a notable advantage in terms of usability when compared to TensorFlow. The TensorFlow version which we used in 2.12.0



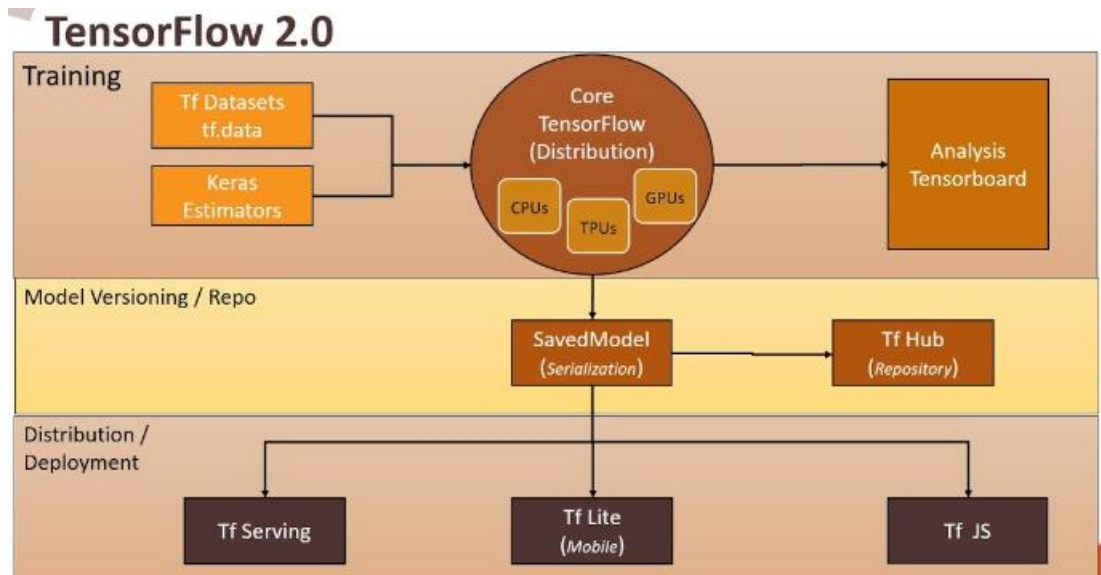


Fig 12. TensorFlow architecture

Here we are implementing VGG 16, VGG 19, Resnet 50 and Resnet 101 in Google colab pro with V100 GPU using python code.

### 3.6.1 Convolutional neural network (CNN):

The convolutional neural network (CNN) is an algorithm and model used in deep learning that accumulates knowledge directly from data. CNN have shown to be quite useful in the realm of image analysis due to their ability to discover subtle patterns, aiding the classification of numerous classes and their groupings. CNN comprises of numerous layers, wherein each layer is designed to acquire the ability to recognize distinct features within an image. In the context of image processing, filters are employed to perform convolutions on training images at various resolutions. The resulting images are subsequently utilized as input for the subsequent layer in the network architecture. In comparison to other deep learning methods, the current methodology requires little prior data preprocessing.

This proposed technology exhibits a high degree of compatibility with automated vision tasks, providing it particularly well-suited for applications that necessitate robust object recognition capabilities. Examples of such applications includes skin, breast cancers and facial recognition systems. It is capable of extracting critical insights from both temporal and visual data. CNN neurons follow the same organizational pattern as the frontal lobe of the human brain, which is primarily responsible for the complicated interpretation of visual data.

Deep learning is an area of machine learning that attempts to replicate the human brain's ability to process and evaluate data. It entails training artificial neural networks using massive datasets. The CNN framework is divided into three layers: a convolutional layer, a pooling layer, and a fully connected (FC) layer. The convolutional layer is the primary tier in the neural network design.

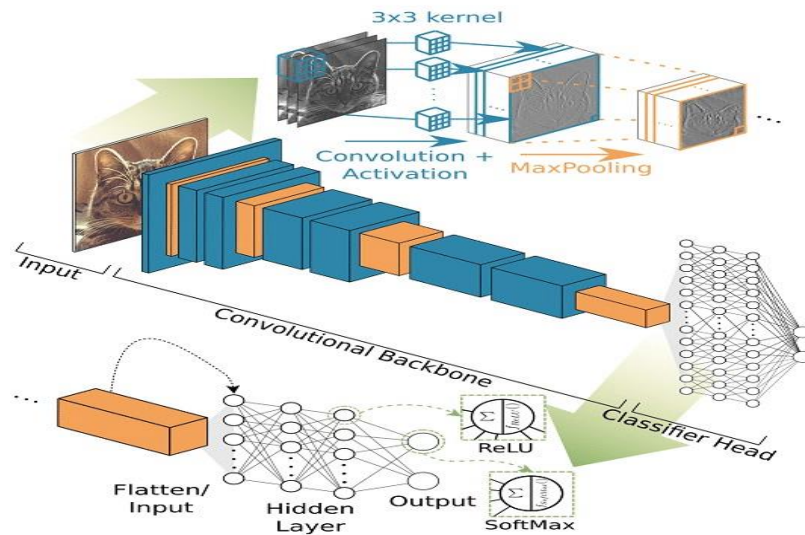


Fig 13. CNN Architecture

For effective training, convolutional neural networks (CNNs) need a large amount of data. Collecting an extensive set of medical photos is a difficult task. In situations where the size of the database is limited, transfer learning and fine tuning have acquired significant recognition.

The utilization of convolution in computer vision research is underpinned by three fundamental concepts that have served as key motivators for researchers in this field. These concepts include sparse interaction, parameter sharing, and equivariant representation.

The pooling layer's job is to replace the network's output in certain places by figuring out a summary figure based on the outputs that are close. This process makes it easier to reduce the size of the image in space, which reduces the amount of work that needs to be done on it and its weight. The pooling process is done in a different way on each slice of the representation.

The final convolution or pooling layer's output feature maps are often compressed to a one-dimensional (1D) array of integers that is linked to one or more complex layers, in which every data point is related to every output. The features captured by the convolution layers and down.

sampled by the pooling layers are mapped to the network's final outputs, such as classification probabilities, by a selected set of the layers that are fully connected. Each fully linked layer is followed by a function that is nonlinear such as ReLU.

All negative numbers in ReLU are turned to zero, merely removing the presence of any negative values. The Vanishing Gradient problem does not pose an issue in cases where the Maximum Threshold values are set to Infinity. Consequently, the output prediction accuracy and efficiency are maximized. The speed of the activation function is notably higher when compared to alternative activation functions.

In the proposed system, further we discuss about VGG-16, VGG-19, Resnet-50 and Resnet-101.



### 3.6.2 VGG-16

In our experiment, we first implement a convolutional neural network model, specifically a VGG-16 that had already been pre-trained. Due to the small size of our image dataset (which has 963 images), we had to make adjustments to the model to avoid the risk of overfitting by keeping just a small percentage of the features.

There are 16 convolutional layers in the VGG-16 model, and the filter size is 3\*3. The initial node of the network has the coordinates 224, 224, and 3. Here, 224\*224 represents the height and width measurement of the input image in pixels and 3 represents the colour in input image i.e., RGB (red, green and blue).

In addition, the network uses max pooling in five stages, each of which has a grouping size of 2\*2. However, a SoftMax output layer binds the two layers together at the deepest possible level. The VGG-16 model stands out due to its vastness, which is over 138 million parameters. Deep neural networks are created with this technique by stacking multiple convolutional layers. The network's capacity to develop and extract implicit characteristics is considerably enhanced by this technique. Vgg-16 is used as the feature extractor in many fields like medical diagnosis, object detection, video analysis, face recognition, robots etc.

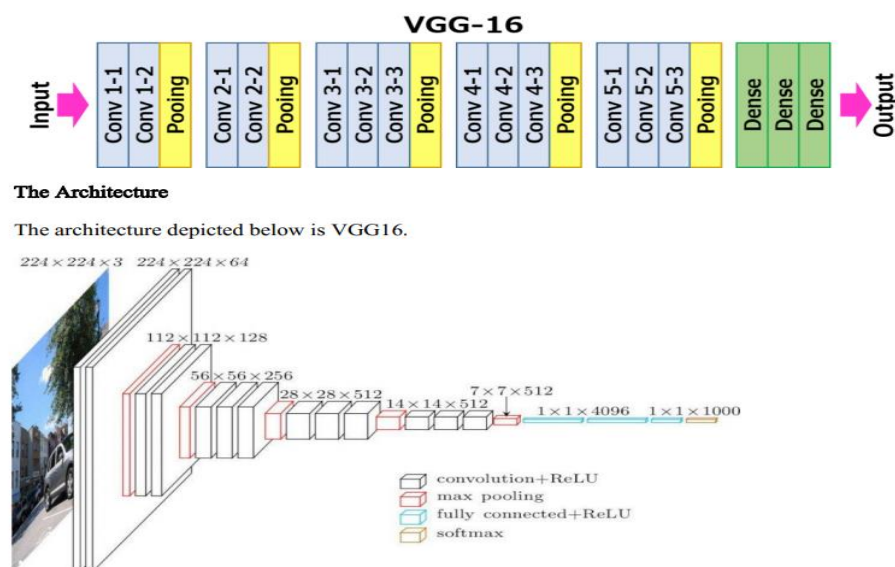


Fig 14. VGG-16 Architecture.

### 3.6.3 VGG-19

The VGG-19 standard is an enhanced version of the VGG-16 standard, incorporating subtle adjustments. This network is comprised of 19 distinct layers. There is an option of loading a network that has already been trained using the database, which contains over one million images. Images are able to be categorized into one thousand distinct object types (animals, tree, building, food, etc) because of the pretrained network. Since it uses a large number of filters with a dimension of 3 x 3 in every convolutional layer, it has generated a lot of interest as a method that is effective for the classification of images.

VGG-19 demonstrates that there are sixteen convolutional layers being utilized for feature extraction, and the following three layers are being utilized for categorization. The layers that are utilized for feature extraction are broken up into five blocks, and a max-pooling layer comes after each of those groups in turn. This model takes an image with dimensions of  $224 \times 224$  as its input, and it returns the description of the object that can be seen in the image as its output. The image in examining is a 3-channel image, where the average RGB value has been reduced. It has continuously attained higher levels of performance in contrast to other models in image classification and identification, object detection. We were successful in training a convolutional layer to perform a particular classification task by utilizing both full connected layer and a SoftMax layer as part of our data-driven approach.

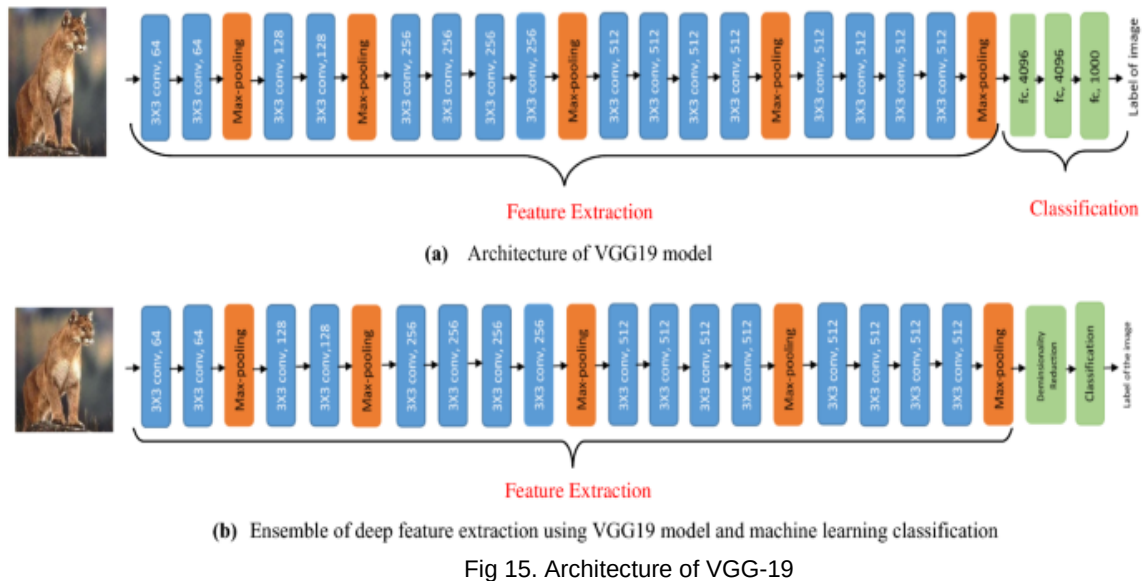


Fig 15. Architecture of VGG-19

### 3.6.4 ResNet

The fundamental concept behind ResNet is that each successive layer of the architecture is able to acquire new knowledge from hidden functions by referring to the layer directly below it in the hierarchy. The design can be easily adjusted and will experience a notable increase in accuracy as a result of this. Within the scope of this study, we made use of pre-trained models of two distinct versions, namely ResNet-50 and ResNet-101.

### 3.6.5 ResNet-50

A variation of the ResNet model known as ResNet50 contains 48 Convolution layers in addition to a MaxPool layer and an average Pool layer. The computational task involves a significant number of floating-point operations, specifically  $3.8 \times 10^9$  (since it has 50 deep network layers). In this ResNet50 architecture, a single layer is constructed by applying a convolution operation with a kernel size of  $7 \times 7$ . This convolution is performed using 64 distinct kernels, each with a stride size of 2. Next, we observe the utilization of max pooling, where it also uses a stride size of 2.

ResNet50 made use of a stack with three layers, instead of two layers. As a result, each of the 2-layer blocks that were used in Resnet34 were changed to a 3-layer bottleneck

block, and the resulting design was called Resnet 50. This model's accuracy outperforms that of the 34-layer ResNet model by a significant margin because of image classifications quality in it. A performance of 3.8 billion FLOPS is achieved by the 50-layer ResNet architecture. At the very end of the classification process, SoftMax is used as the final layer. It can be utilized on all the computerized vision tasks like object identification, self-driving cars, etc.

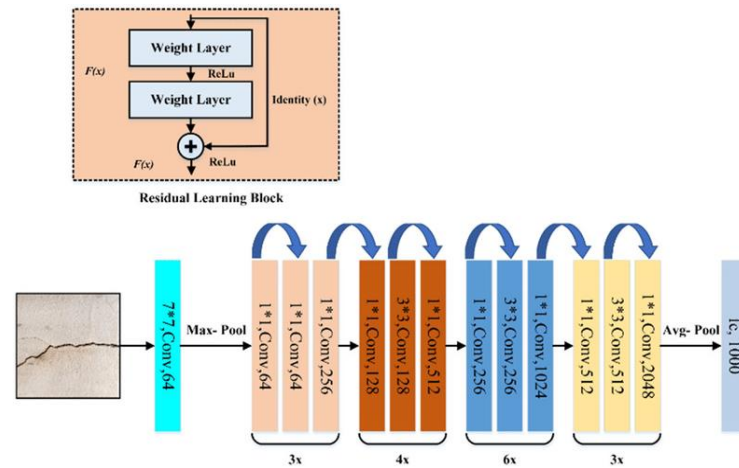


Fig 16. ResNet- 50 Architecture

### 3.6.6 ResNet-101

ResNet-101 represents a notable iteration of the ResNet architecture, claiming an impressive depth of 101 layers. This increased layer count is a contributing factor to its improved efficacy, which allows it to surpass the capabilities of its earlier predecessors. The fundamental component of ResNet is the residual block, comprising of two or three convolutional layers along with a collection of unique shortcut links. The skip connection serves as the central element of the residual block, facilitating the direct addition of the input parameter to the end result of the convolutions. The neural network acquires knowledge of the residual, which refers to the disparity between the input and output of the convolution layers.

The ResNet-101 architecture strategically integrates bottleneck blocks (The input from the previous layer is convolutionally processed and appended to the output via a skip connection. Information can pass over the block unaffected) within its deeper layers to optimize computational resources and enhance the network's overall efficiency. This approach effectively reduces the processing load while maintaining the network's performance. Within the bottleneck block, there exist three distinct convolutional layers. The dimensions of the objects are 1x1, 3x3, and 1x1, respectively. The utilization of 1x1 convolutions involves an initial reduction in the number of channels, followed by subsequent augmentation of the channel count. The process of feature extraction is carried out by the 3x3 convolution.

ResNet101 uses ReLu to achieve the non-linearity. In order to improve the consistency of the training process, batch normalization is implemented as a technique of standardizing the input for each layer.

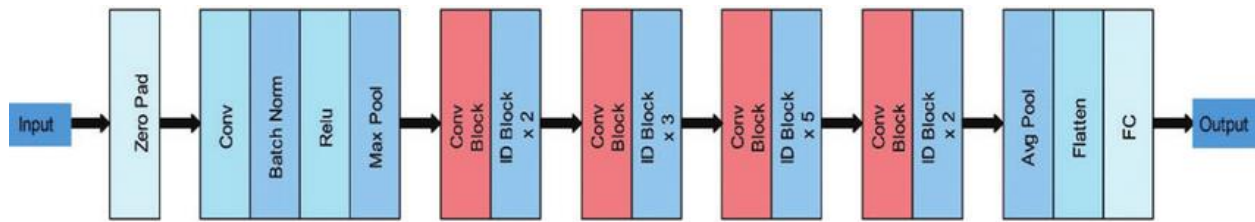


Fig 17. Architecture of ResNet-101.

### 3.6.7 Ensemble classifier

The outcomes of machine learning can be enhanced by means of ensemble learning by merging multiple different models. When compared to using just one model, this strategy enables the generation of much more accurate predicted performance. The fundamental concept is to acquire a set of classifiers and then to choose which classifiers is the best.

In this study, we assess the performance of four deep learning models: Vgg-16, Vgg-19, Resnet-50, and Resnet-101. The evaluation was conducted to determine the most effective models, which were merged in order to create an ensemble classifier. The ensemble classifier will be generated based on the accuracy they get.

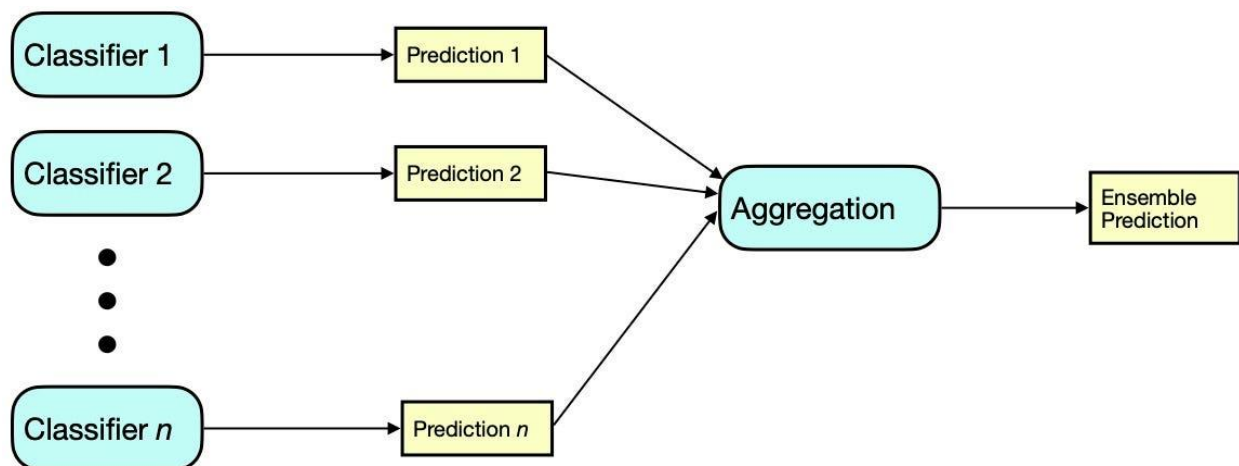


Fig 18. Ensemble classifier basic architecture.

## 4 Analysis

The findings from the research study on the multi-class cervical cancer diagnosis using liquid-based cytology (LBC) Pap smear images are included in this part. The study's objective was to develop an accurate and efficient classification algorithm that could distinguish between normal, precancerous, and cancerous cervical cells. The datasets were gathered and prepared, and the process also comprised feature extraction, model selection, model training and evaluation, and performance evaluation. This chapter presents a detailed analysis and discussion of the obtained results with emphasis on the performance of the created model, its benefits and drawbacks, and the implications for the diagnosis of cervical cancer.

The results of the study demonstrate the potential of employing LBC Pap smear images for several cervical cancer classifications. In identifying healthy, precancerous, and cancerous cervical cells, the classification model showed great promise and good accuracy. The study advances the field's body of knowledge and increases the likelihood of earlier cervical cancer detection and improved patient outcomes. Additional research and validation are needed to improve and optimize the model's performance while taking into consideration different populations and clinical scenarios.

The findings will be analysed to draw meaningful inferences, assuming that the trained models make correct predictions. It is believed that the performance measures for each model will show how well it succeeds in correctly identifying cervical cancer cases based on LBC Pap smear pictures. By analysing the performance of multiple models, insights into the prospects of using computer vision and machine learning methodologies including CNN, VGG-16, VGG-19, ResNet-50, and ResNet-101 for multi-class cervical cancer diagnosis will be gained.

The findings will be interpreted in light of the assumptions made, the body of literature previously available, and the research objectives, providing meaningful information on the practicality and success of this technique.

To improve the quality and consistency of the photos, preprocessing techniques such image scaling, normalization, noise reduction, and artefact removal were used. The findings demonstrated that scaling images helped standardize their proportions, allowing for quick processing. Image clarity improved as a result of noise reduction. In order to obtain morphological, statistical, and textural information, the pre-processed photographs were utilized.

According to the findings, these traits, accurately captured the structural and textural properties of the cells. The statistical analysis discovered significant differences in feature values between the various classes, confirming the discriminatory power of the obtained features.

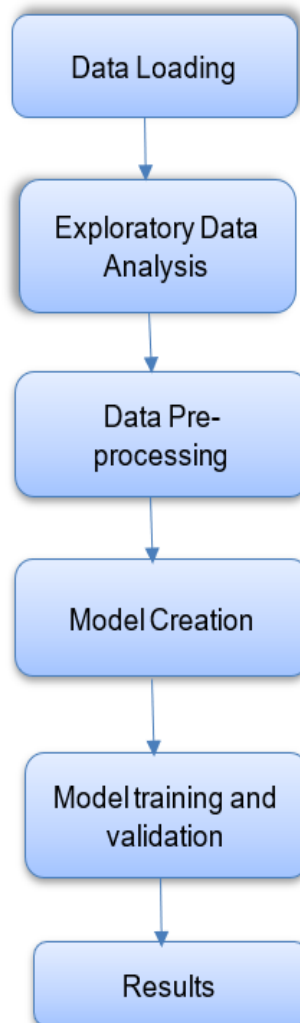
Convolutional neural networks (CNN), support vector machines (SVM), and random forests were among the machine learning algorithms that were examined for model selection. Cross-validation methods were used to evaluate each algorithm's performance. The CNN-based model beat the other algorithms, according to the results, attaining the highest accuracy score. The CNN performed better than other systems in detecting cervical cancer because of its capacity to learn intricate patterns and hierarchical representations from the images.

The research study showed an encouraging outcome for the multi-class identification of cervical cancer using LBC Pap smear images and a computer-aided diagnostic (CAD) system. The CAD system demonstrated a high level of accuracy in classifying the images into various stages of cervical cancer after being trained on a subset of expert-annotated images. Accuracy score is used to assess the system's performance. The findings suggested that the CAD approach may help cytopathologists identify cervical cancer by offering quicker and perhaps more precise evaluations of LBC Pap smear pictures (Jain, et al., (2022)). The CAD system sorts the photos into specified classes by extracting pertinent information from the images using cutting-edge image processing techniques and machine learning algorithms. This may increase the diagnostic procedure's effectiveness, resulting in earlier detection and better patient outcomes.

In conclusion, numerous crucial phases are included in the data analysis phase of this work, including feature extraction, model training and evaluation, and result interpretation. This study seeks to gain understanding and reach meaningful conclusions regarding the use of LBC technique, machine learning techniques, CNN, (VGG-16,19), (ResNet-50,101). These conclusions are based on assumptions about the applicability and quality of the data, the characteristics and models used, and the validity of the results which helps in enhancing patient outcomes and healthcare practices.

## 5 Design

This case study intends to evaluate the performance of traditional Pap smear and liquid-based cytology (LBC) in the multi-class diagnosis of cervical cancer. Benchmark dataset is used in this project which has total of 963 images collected from 460 patients. The type of data in dataset is purely images. Only a few of the crucial demographic and clinical information that was gathered. The findings of LBC and conventional Pap smears' cytology tests were evaluated and classified using the Bethesda system across various classes. There is a flow chart designed for this research as shown below:



### 5.1 Data Loading

The initial phase of the processes involves a task of loading and reading the dataset. Its primary function is to load this dataset into Google drive, Mount that folder in to the google colab, check if the dataset is executing properly and conduct exploratory data analysis (EDA). The output produced by this phase offers valuable insights into various aspects of the data, including data characteristics, the arrangement of the data and identification of noise in images. The insights derived from this block serve as input for the subsequent block, which is responsible for data pre-processing.

## 5.2 Exploratory Data Analysis

In this section, we are resizing, normalize the images using pixel scaling method and reducing the noise from the images to get the proper image results for further analysis.

## 5.3 Data Preprocessing

Here, we eliminate if there are any errors that impacts the final model's accuracy. Feature extraction process is done depending on the machine learning models we chooses. This pre-processed data will be split into two parts for training and testing purpose.

## 5.4 Model Creation, training, and execution

In this study, we are constructing models for VGG16, VGG19, ResNet-50, and ResNet-101 architectures. Prior to developing a model, it is imperative to ensure that all the necessary libraries for TensorFlow and keras are installed. All of the models have been constructed and their corresponding code, together with the model summary (), is depicted in the picture provided below.

Each model undergoes training using the pre-trained dataset, specifically ImageNet. This training process involves the addition of a maxpooling layer, batch normalization, and the activation of ReLu by the inclusion of a fully connected layer consisting of 128 units. The dropout function is employed as a means of mitigating the issue of overfitting. Additionally, the ADAM optimizer is employed. The early stopping function is employed to train the model effectively by mitigating overfitting and determining the ideal training breakpoint without requiring the execution of all epochs.

## 5.5 Results

This step mainly focuses on plotting graph for each model's accuracy and loss also, Tensor-board visualization tool is used for displaying the timing and accuracy of each epoch.

Based on the all the process, the accuracy and loss for each model is finely tuned and compared to see which is the better model that separates the cancer cell based on its stage of cancer which helps in early detection of the disease and provide treatment to patients on time.



## 6 Implementation

### 6.1 Platform used for coding

The programming language employed in this project is Python. The Google Colab is the text editor that is being utilized here. In this case, Python is utilized both in the construction of the neural network and in the implementation of the models that are utilized for the classification of the image dataset that has been taken for this project.

Here, we have utilized a benchmark dataset (refer to chapter 3.1) for the purpose of classifying and detecting the cervical cancer cells.

Python is an open-source programming language that offers a convenient means of implementing various methodologies and important technologies.

Python, a widely used programming language, offers an extensive range of libraries that greatly enhance its functionality. Notable among these few libraries are TensorFlow, Keras, Pandas, NumPy, Matplotlib, and Seaborn.

### 6.2 Google collab

The Google Colaboratory, more frequently referred to as the Google Colab, is a cloud-based research application that has been created by Google. It is identical to Jupiter notebook in that it enables users to construct and run Python programs in their web browser. Since the structure of the code editor is organized cell by cell, it is possible for the author to view the output of each line of code once it has been executed. POSIX is the name of the computer operating system that is utilized by Colab.

The primary allure of employing this software lies in the GPU capabilities provided by Colab Pro. The Colab Pro computing platform is equipped with a hard disc capacity of 34 gigabytes, while its central processing unit (CPU) is an Intel Xenon operating at a clock speed of 2.30 gigahertz. The utilization of Colab Pro enables users to maintain a continuous connection to the notebook for a duration of 24 hours, thereby mitigating any concerns regarding timeouts. This particular feature is highly beneficial in nature.

When training a model with the images in a neural network, these images need to be translated into matrices with the pixel colours and intensity levels, and the numerical values of these matrices will be the number of arithmetical operations. This is necessary so that the model can be trained. Because of this, a significant amount of processing capacity from the GPU is required. Colab utilizes a powerful GPU known as the TESLA P100-PCIE-16GB in order to address issues relating to computer vision. Colab comes equipped with CUDA out of the box due to the fact that the GPU is an NVIDIA product. The primary memory that comes standard with Colab Pro is 26.30 gigabytes, which is suitable for solving 99 percent of all machine learning issues.

### 6.3 Model Building

After completing the steps necessary to prepare the dataset and carry out the pre-processing operations, one of the most important goals in the process of constructing the model was to prepare the environment in the Colab. The fact that many of the dependencies had been installed in Colab at an earlier time. Because of this, the amount of time necessary to install the dependencies was cut down. Another essential step was taking simply to organize the folders and paths utilized by the TensorFlow object identification API. It is necessary to finish all of these activities before proceeding with the model's training and verification.

### 6.4 Handling the Data

The dataset has been submitted to Google Drive in the form of two distinct compressed files. The subsequent phase entailed the integration of Google Drive into the pre-existing Colab infrastructure. The initial step involves the importation of the Google Drive dependency, which is subsequently followed by the execution of the relevant cell until a request for authentication that is made by Google. The achievement of this objective can be realized by adhering to the subsequent procedures. Upon removal of the ".zip" extensions, the files contained within this location can be subsequently accessed and opened. The data was divided using a 9:1 split, allocating 90% of the time for training purposes and reserving the remaining 10% for testing. The allocation was conducted in a completely arbitrary manner.

```
#splitting training and test data by preserving the class distribution:
train_data, val_data = train_test_split(data, test_size=0.1, random_state=42, stratify=data['class'])
print(train_data['class'].value_counts(dropna=False))
print(val_data['class'].value_counts(dropna=False))
```

```
2    550
0    147
1    102
3     66
Name: class, dtype: int64
2     62
0     16
1     11
3      8
Name: class, dtype: int64
```

Fig.19 Splitting train and test data.

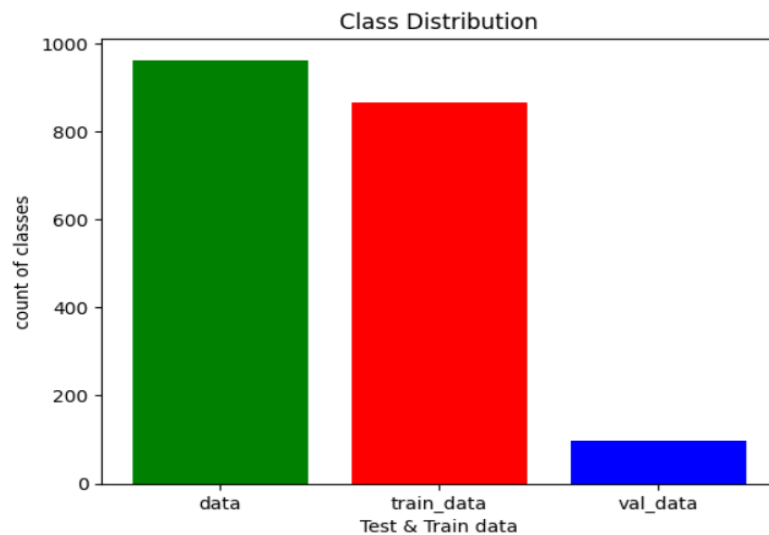


Fig. 20 Graphical presentation of test and train data.

## 6.5 Model Training

In order to facilitate the training of the model, the standard TensorFlow and keras libraries are used for training and testing in the python script, where it would be utilized for the pre-processing steps of the dataset. Subsequently, the images were subjected to a process of reconfiguration in order to align them with the corresponding models. The dataset comprises four distinct categories of images, each of which is associated with class labels ranging from 0 to 3.

Here, initially we will implement the VGG16 model. Since, there is a good accuracy rate for the first 15 epochs from the VGG16 model, we did not proceed for next one from which we will choose the highest epoch's accuracy and it is similar for all the other models like VGG-19, ResNet-50 and ResNet-101 also, for all the models the graphical representation of accuracy and loss is drawn out.

```
#assigning labels to class:
change = {
    'High squamous intra-epithelial lesion' : '0',
    'Low squamous intra-epithelial lesion': '1',
    'Negative for Intraepithelial malignancy' : '2',
    'Squamous cell carcinoma': '3',
}

data['class'] = data['class'].map(change)
data
```

Fig 21. Assigning labels to each class.

|            | <b>filename</b>                                   | <b>class</b> |
|------------|---------------------------------------------------|--------------|
| <b>0</b>   | /kaggle/input/cervical-cancer/Low squamous int... | 1            |
| <b>1</b>   | /kaggle/input/cervical-cancer/Negative for Int... | 2            |
| <b>2</b>   | /kaggle/input/cervical-cancer/Negative for Int... | 2            |
| <b>3</b>   | /kaggle/input/cervical-cancer/Negative for Int... | 2            |
| <b>4</b>   | /kaggle/input/cervical-cancer/Low squamous int... | 1            |
| ...        | ...                                               | ...          |
| <b>957</b> | /kaggle/input/cervical-cancer/High squamous in... | 0            |
| <b>958</b> | /kaggle/input/cervical-cancer/Low squamous int... | 1            |
| <b>959</b> | /kaggle/input/cervical-cancer/Negative for Int... | 2            |
| <b>960</b> | /kaggle/input/cervical-cancer/Negative for Int... | 2            |
| <b>961</b> | /kaggle/input/cervical-cancer/High squamous in... | 0            |

Fig 22. Results for the above code.

## 7 Results

Classification on the benchmark dataset, which is extracted from the Mendeley website, collects the images by utilizing two different pap smear procedures, namely liquid-based cytology and the standard pap test, which are then evaluated.

In this research, a review and investigation of the performance of four different deep learning models made use of transfer learning to leverage the performance. These models were compared and contrasted based on how well they learned. In addition, for the goal of multi-class diagnostic prediction of cervical cells disease, we compared the performance of an ensemble classifier with that of individual models that had been pre-trained sequentially. This was done in order to determine which method produced the most accurate results.

### 7.1 Traditional pap smear vs LBC

The objective of this study was to conduct a comparative analysis of the diagnostic performance of liquid-based cytology (LBC) and standard Pap smears in the context of multi-class cervical cancer diagnosis. In comparison, it is shown that LBC was more sensitive to abnormal instances than traditional Pap smears were. This shows that LBC is a better screening technique for detecting cellular abnormalities and lowering the incidence of false-negative results. By enabling prompt action and earlier diagnosis of pre-cancerous lesions, the greater sensitivity of LBC can improve patient prognosis Bogdanovska and Velickova, et.,al (2018).

The most found pathogens from the research are trichomonas vaginitis, Candida, and Bacterial vaginosis. These infectious pathogens can be accurately predicted and detected by LBC, which helps direct focused treatment strategies and stop illness development. Parity index and cervical anomalies showed how childbearing affects cervical health. The study found that a higher parity index increased cervical abnormalities. Multiple-pregnant women may benefit from more frequent testing and monitoring.

Prospective studies with larger sample numbers and multi-centre partnerships should increase the evidence base. Long-term follow-up studies and cost-effectiveness evaluations are needed to evaluate LBC as a primary screening method.

### 7.2 Multi-class classification and Trained models results

An evaluation was carried out to determine which classifiers from already-existing deep convolutional neural networks (vgg-16, vgg-19, resnet-50, and resnet-101) were most suited for the task that helps in predicting cervical cancer stage using pap-smear images. The model with the best overall performance will be determined by analysing the loss as well as the accuracy, precision and F1 consistency across the training, validation, and testing phases for each of the various models.

| Model         | Test Accuracy | Validation Accuracy |
|---------------|---------------|---------------------|
| 1. VGG-16     | 94%           | 93%                 |
| 2. VGG-19     | 91%           | 86%                 |
| 3. ResNet-50  | 90%           | 78%                 |
| 4. ResNet-101 | 83%           | 78%                 |

Table 2. Accuracy for all the models.

Among all the models, VGG-16 has the highest accuracy of 94% which can be further used for image classification to find the cancer cells.

```

Epoch 1/10
25/25 [=====] - 69s 3s/step - loss: 0.8190 - accuracy: 0.6710 - val_loss: 0.7911 - val_accuracy: 0.6437
Epoch 2/10
25/25 [=====] - 66s 3s/step - loss: 0.5145 - accuracy: 0.7931 - val_loss: 0.7065 - val_accuracy: 0.7816
Epoch 3/10
25/25 [=====] - 77s 3s/step - loss: 0.3903 - accuracy: 0.8496 - val_loss: 0.5732 - val_accuracy: 0.7816
Epoch 4/10
25/25 [=====] - 68s 3s/step - loss: 0.3468 - accuracy: 0.8638 - val_loss: 0.5092 - val_accuracy: 0.7931
Epoch 5/10
25/25 [=====] - 71s 3s/step - loss: 0.2966 - accuracy: 0.8792 - val_loss: 0.4458 - val_accuracy: 0.8046
Epoch 6/10
25/25 [=====] - 80s 3s/step - loss: 0.2493 - accuracy: 0.9023 - val_loss: 0.3762 - val_accuracy: 0.8391
Epoch 7/10
25/25 [=====] - 66s 3s/step - loss: 0.2076 - accuracy: 0.9229 - val_loss: 0.2694 - val_accuracy: 0.8966
Epoch 8/10
25/25 [=====] - 65s 3s/step - loss: 0.2262 - accuracy: 0.9229 - val_loss: 0.2704 - val_accuracy: 0.8966
Epoch 9/10
25/25 [=====] - 64s 3s/step - loss: 0.2058 - accuracy: 0.9177 - val_loss: 0.2348 - val_accuracy: 0.8851
Epoch 10/10
25/25 [=====] - 65s 3s/step - loss: 0.1601 - accuracy: 0.9422 - val_loss: 0.1828 - val_accuracy: 0.9310
<keras.callbacks.History at 0x7a7865bdc1f0>

```

Fig 23. VGG-16 model's test and validation accuracy

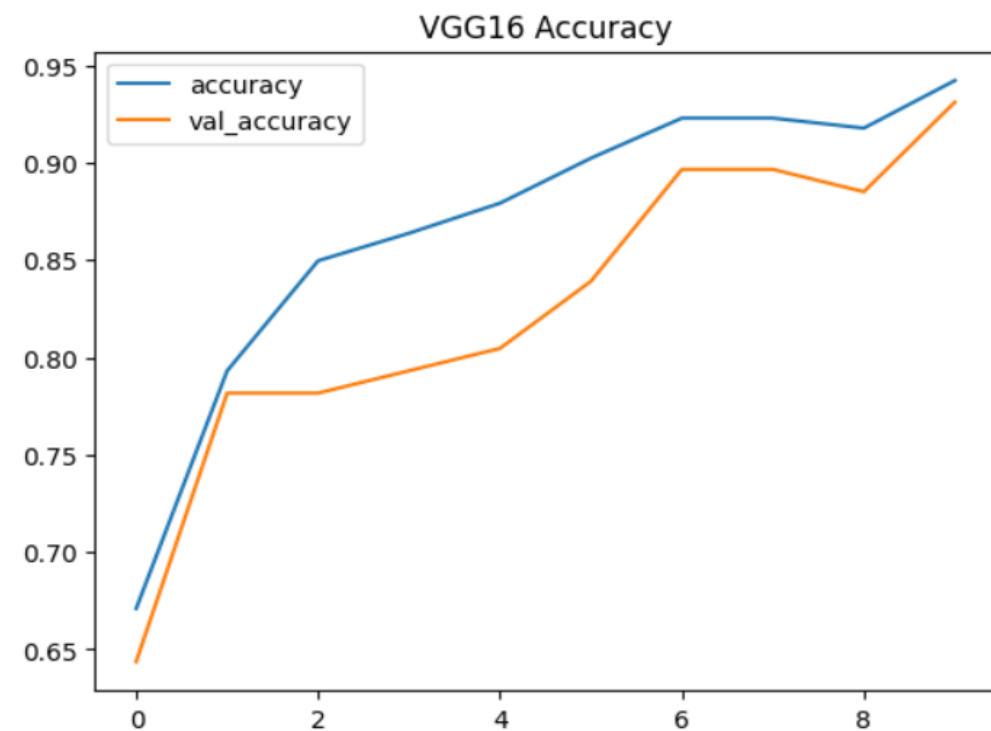


Fig 24. Accuracy of VGG-16

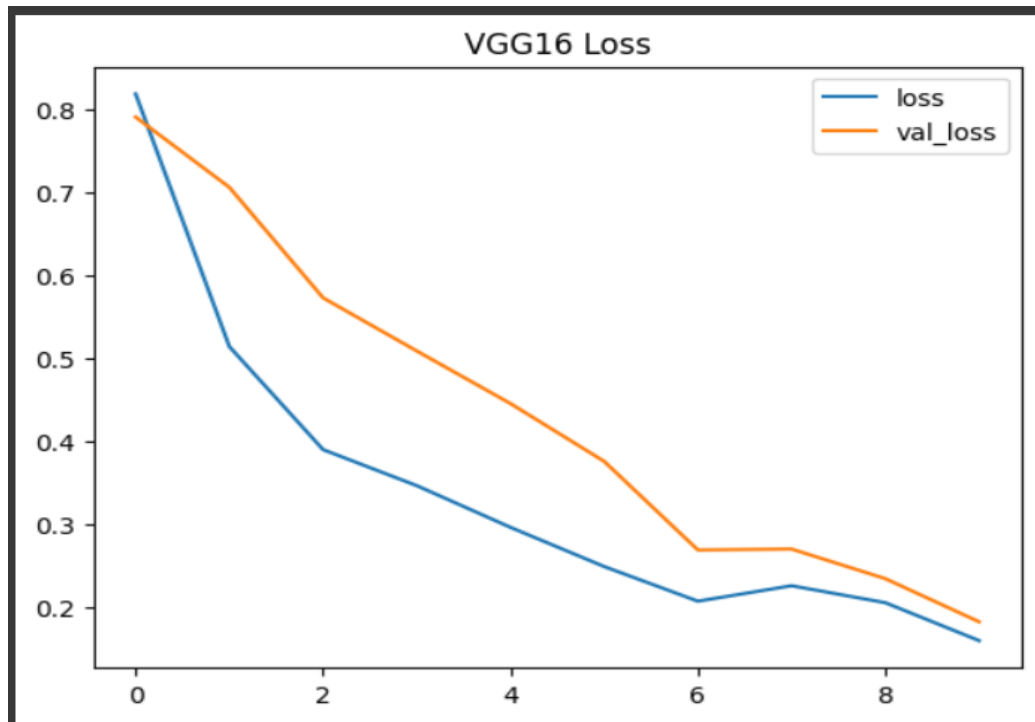


Fig 25. Loss for VGG-16

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.75      | 1.00   | 0.86     | 9       |
| 1            | 1.00      | 0.75   | 0.86     | 8       |
| 2            | 0.98      | 1.00   | 0.99     | 43      |
| 3            | 1.00      | 0.50   | 0.67     | 4       |
| accuracy     |           |        | 0.94     | 64      |
| macro avg    | 0.93      | 0.81   | 0.84     | 64      |
| weighted avg | 0.95      | 0.94   | 0.93     | 64      |

Fig 26. Values of VGG-16 model factors.

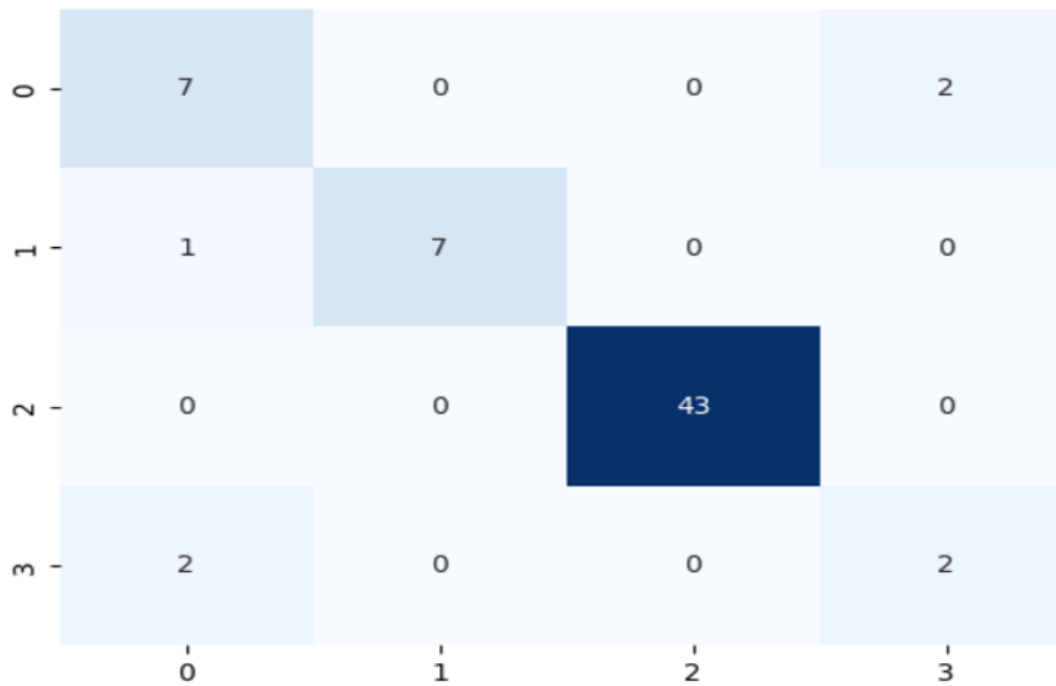


Fig 27. Confusion Matrix of VGG-16

Here, total number of images are 64 and they are distributed as shown above where, rows represent predicted images count and columns represent actual images count.

Below are the classes for each type of cancer:

Class 0 – High squamous intra-epithelial lesion

Class 1 - Low squamous intra-epithelial lesion

Class 2 – Negative for Intra-epithelial malignancy

Class 3 – Squamous cell carcinoma

- For Class 0, the correctly predicted images are 7 and wrongly predicted(or) misclassified with class 3 are 2 images.
- Class 1 - Correct prediction: 7, misclassification: 1
- Class 2 – Correct prediction: 43, misclassification: 0
- Class 3 – Correct Prediction: 2, misclassification: 2.

For visualization purpose, Tensor-Board is used. Here are the results below for test, validation, and testing:



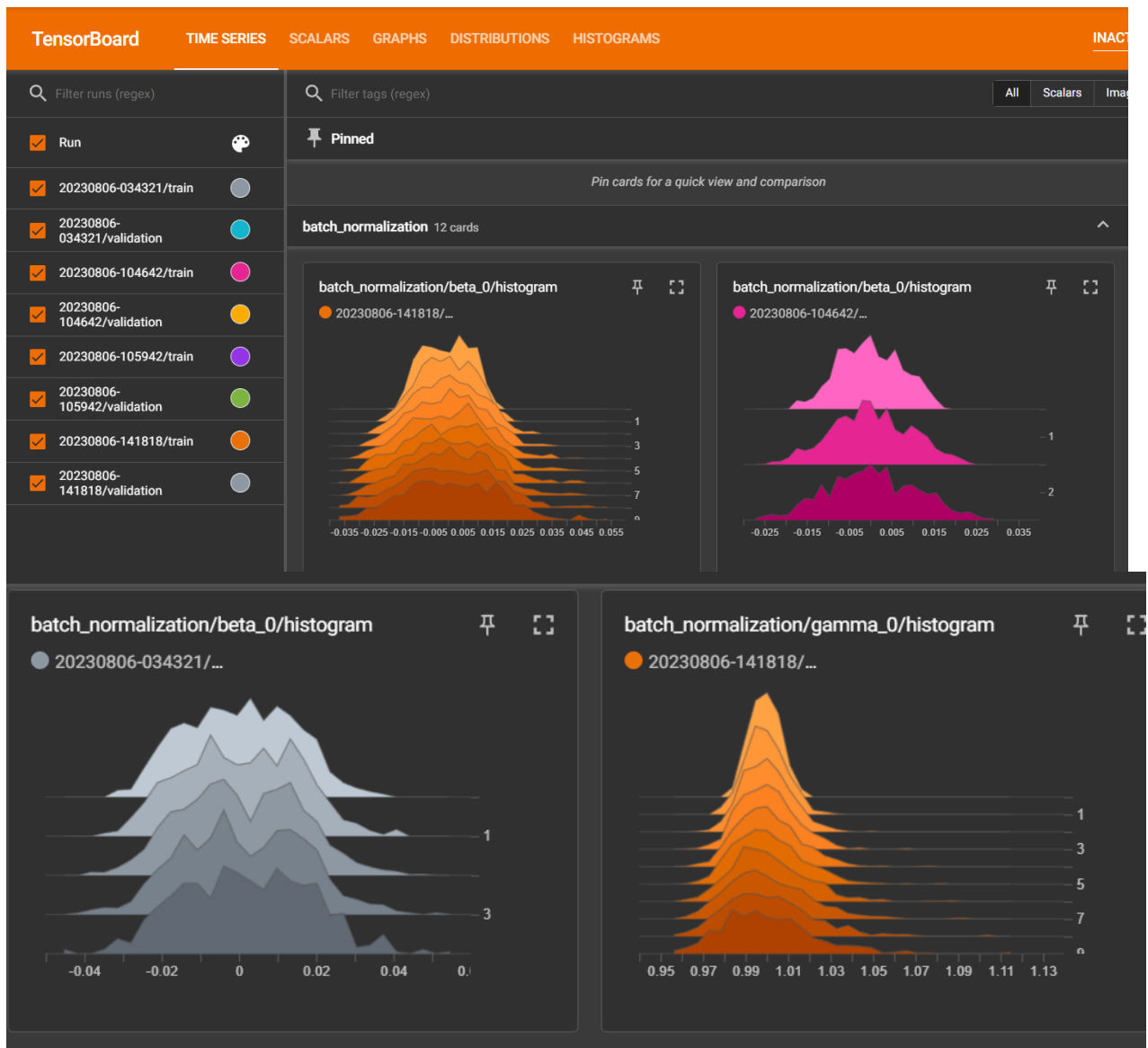


Fig 28. Tensor-Board Visualization for train, test and validation data

```
Epoch 1/10
25/25 [=====] - 72s 3s/step - loss: 0.5185 - accuracy: 0.7879 - val_loss: 1.1299 - val_accuracy: 0.3563
Epoch 2/10
25/25 [=====] - 69s 3s/step - loss: 0.4255 - accuracy: 0.8342 - val_loss: 0.9046 - val_accuracy: 0.5862
Epoch 3/10
25/25 [=====] - 69s 3s/step - loss: 0.3619 - accuracy: 0.8483 - val_loss: 0.8390 - val_accuracy: 0.5747
Epoch 4/10
25/25 [=====] - 68s 3s/step - loss: 0.3231 - accuracy: 0.8766 - val_loss: 0.7785 - val_accuracy: 0.6092
Epoch 5/10
25/25 [=====] - 68s 3s/step - loss: 0.2685 - accuracy: 0.9023 - val_loss: 0.5958 - val_accuracy: 0.7816
Epoch 6/10
25/25 [=====] - 68s 3s/step - loss: 0.2386 - accuracy: 0.9062 - val_loss: 0.5120 - val_accuracy: 0.8046
Epoch 7/10
25/25 [=====] - 68s 3s/step - loss: 0.2301 - accuracy: 0.9139 - val_loss: 0.4243 - val_accuracy: 0.8391
Epoch 8/10
25/25 [=====] - 69s 3s/step - loss: 0.2590 - accuracy: 0.8946 - val_loss: 0.4303 - val_accuracy: 0.8046
Epoch 9/10
25/25 [=====] - 72s 3s/step - loss: 0.2052 - accuracy: 0.9190 - val_loss: 0.3372 - val_accuracy: 0.8621
Epoch 10/10
25/25 [=====] - 67s 3s/step - loss: 0.2405 - accuracy: 0.9126 - val_loss: 0.3077 - val_accuracy: 0.8621
<keras.callbacks.History at 0x7bc86612e620>
```

Fig 29: Epoch run for VGG19's train and validation's accuracy and loss.

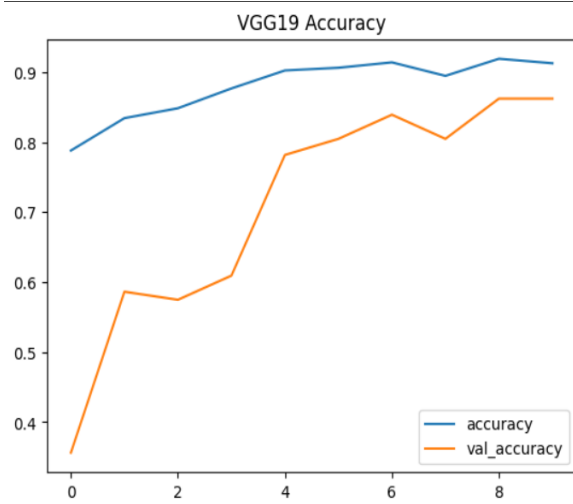


Fig 30. VGG19 Accuracy

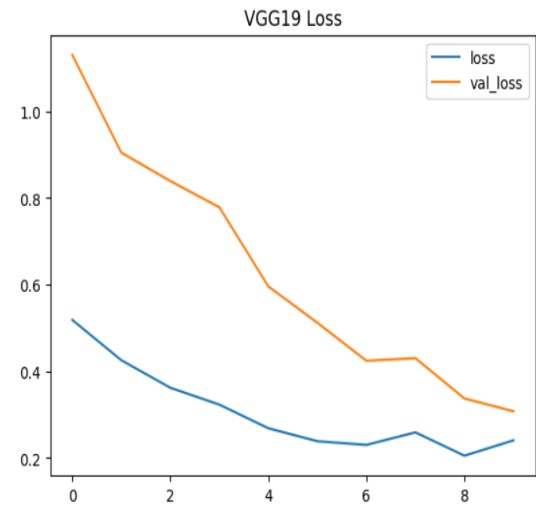


Fig 31. VGG19 Loss

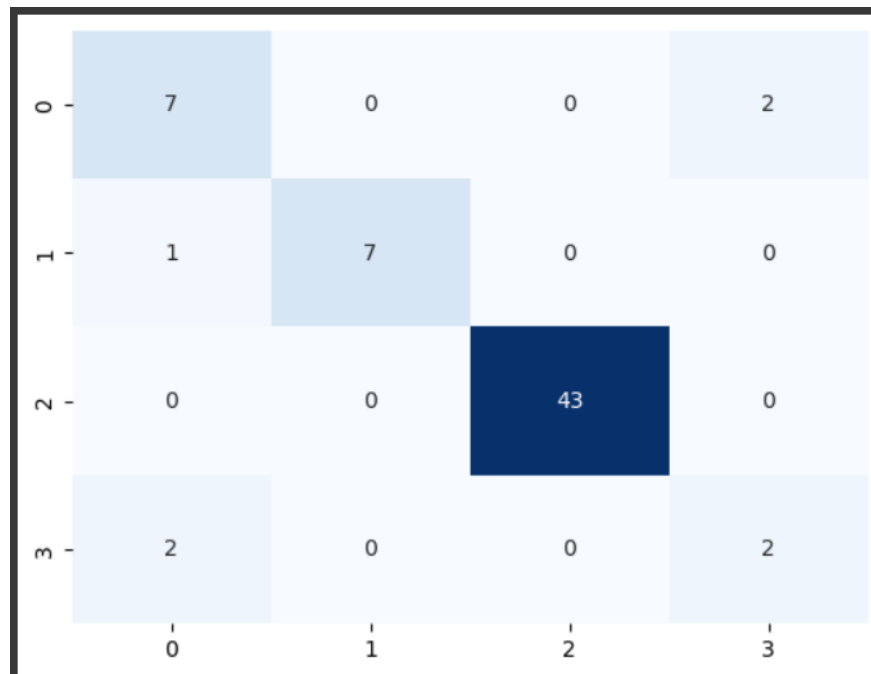


Fig 32. Confusion matrix for VGG19

```

Epoch 1/10
25/25 [=====] - 72s 3s/step - loss: 0.4705 - accuracy: 0.8278 - val_loss: 1.4502 - val_accuracy: 0.5747
Epoch 2/10
25/25 [=====] - 68s 3s/step - loss: 0.4054 - accuracy: 0.8368 - val_loss: 1.3069 - val_accuracy: 0.5172
Epoch 3/10
25/25 [=====] - 91s 4s/step - loss: 0.4316 - accuracy: 0.8445 - val_loss: 1.4390 - val_accuracy: 0.3908
Epoch 4/10
25/25 [=====] - 110s 4s/step - loss: 0.3876 - accuracy: 0.8406 - val_loss: 1.0567 - val_accuracy: 0.6667
Epoch 5/10
25/25 [=====] - 71s 3s/step - loss: 0.3962 - accuracy: 0.8496 - val_loss: 0.9438 - val_accuracy: 0.7011
Epoch 6/10
25/25 [=====] - 87s 4s/step - loss: 0.3483 - accuracy: 0.8522 - val_loss: 0.8748 - val_accuracy: 0.7126
Epoch 7/10
25/25 [=====] - 68s 3s/step - loss: 0.4152 - accuracy: 0.8483 - val_loss: 0.7406 - val_accuracy: 0.7011
Epoch 8/10
25/25 [=====] - 78s 3s/step - loss: 0.3484 - accuracy: 0.8573 - val_loss: 0.6188 - val_accuracy: 0.7241
Epoch 9/10
25/25 [=====] - 69s 3s/step - loss: 0.3259 - accuracy: 0.8728 - val_loss: 0.4998 - val_accuracy: 0.7816
Epoch 10/10
25/25 [=====] - 68s 3s/step - loss: 0.2693 - accuracy: 0.9049 - val_loss: 0.4779 - val_accuracy: 0.7816
<keras.callbacks.History at 0x7a780ad5b4c0>

```

Fig 33. Epoch run for ResNet50's train and validation's accuracy and loss.

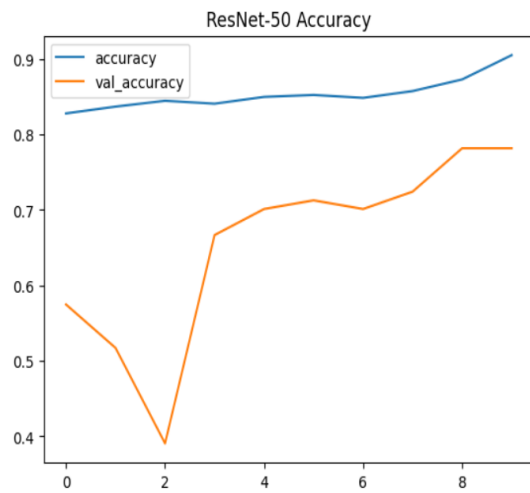


Fig 34. ResNet50 Accuracy

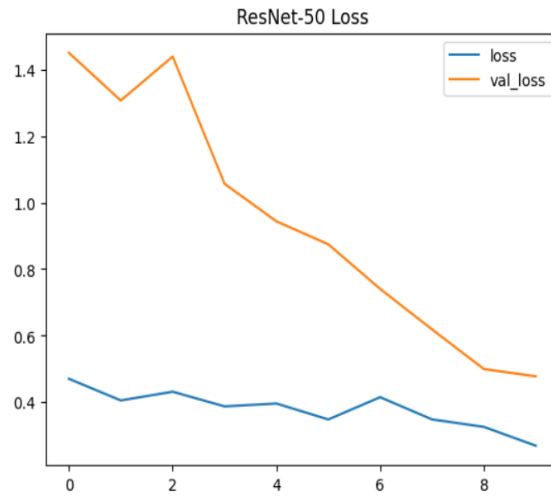


Fig 35. Resnet50 loss

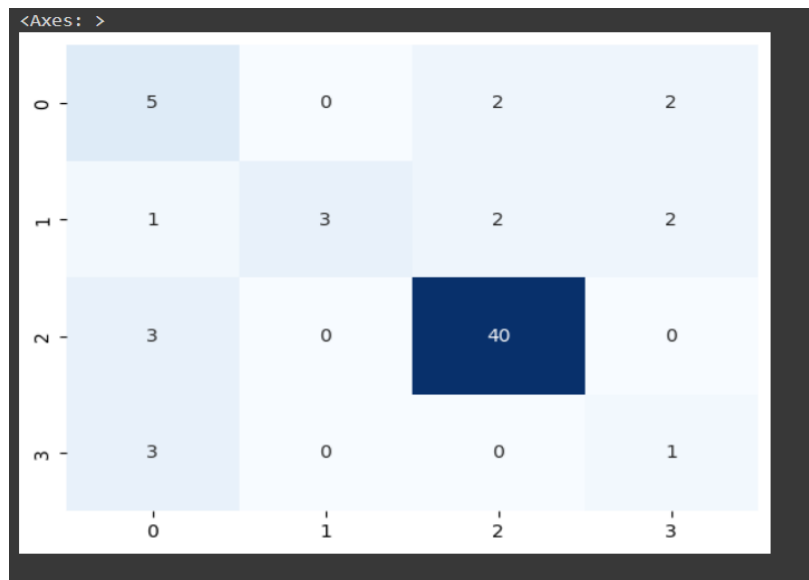


Fig 36. Confusion matrix for ResNet50

```

Epoch 1/10
25/25 [=====] - 90s 4s/step - loss: 0.6089 - accuracy: 0.7571 - val_loss: 2.4685 - val_accuracy: 0.6322
Epoch 2/10
25/25 [=====] - 78s 3s/step - loss: 0.5502 - accuracy: 0.7584 - val_loss: 3.1540 - val_accuracy: 0.6322
Epoch 3/10
25/25 [=====] - 74s 3s/step - loss: 0.5497 - accuracy: 0.7789 - val_loss: 2.3642 - val_accuracy: 0.6322
Epoch 4/10
25/25 [=====] - 76s 3s/step - loss: 0.5265 - accuracy: 0.7725 - val_loss: 2.1179 - val_accuracy: 0.6322
Epoch 5/10
25/25 [=====] - 77s 3s/step - loss: 0.5069 - accuracy: 0.7892 - val_loss: 1.2738 - val_accuracy: 0.6322
Epoch 6/10
25/25 [=====] - 76s 3s/step - loss: 0.4621 - accuracy: 0.8290 - val_loss: 1.0390 - val_accuracy: 0.6552
Epoch 7/10
25/25 [=====] - 76s 3s/step - loss: 0.4374 - accuracy: 0.8239 - val_loss: 0.9638 - val_accuracy: 0.6667
Epoch 8/10
25/25 [=====] - 80s 3s/step - loss: 0.4604 - accuracy: 0.8136 - val_loss: 0.6152 - val_accuracy: 0.7816
Epoch 9/10
25/25 [=====] - 76s 3s/step - loss: 0.4233 - accuracy: 0.8303 - val_loss: 0.5533 - val_accuracy: 0.7931
Epoch 10/10
25/25 [=====] - 74s 3s/step - loss: 0.4087 - accuracy: 0.8329 - val_loss: 0.6131 - val_accuracy: 0.7816
<keras.callbacks.History at 0x7a7790512170>

```

Fig 37. Epoch run for ResNet101's train and validation's accuracy and loss.

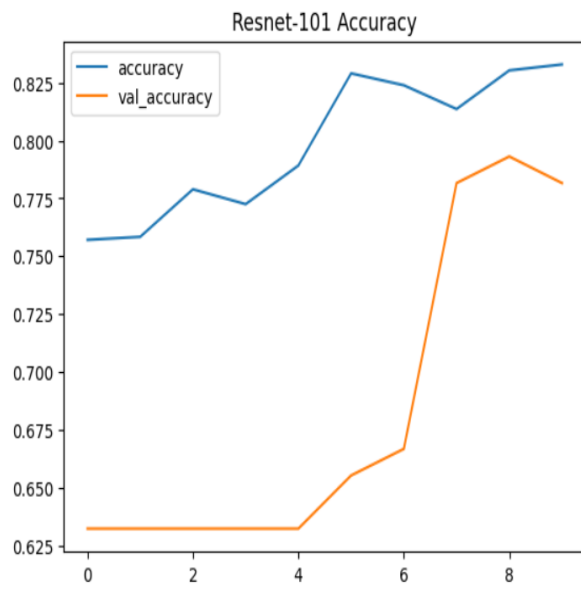


Fig 38. ResNet101 Accuracy

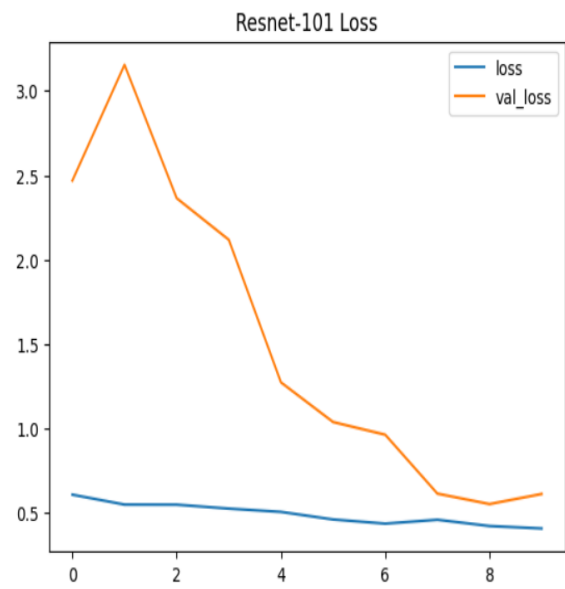


Fig 39. ResNet101 Loss

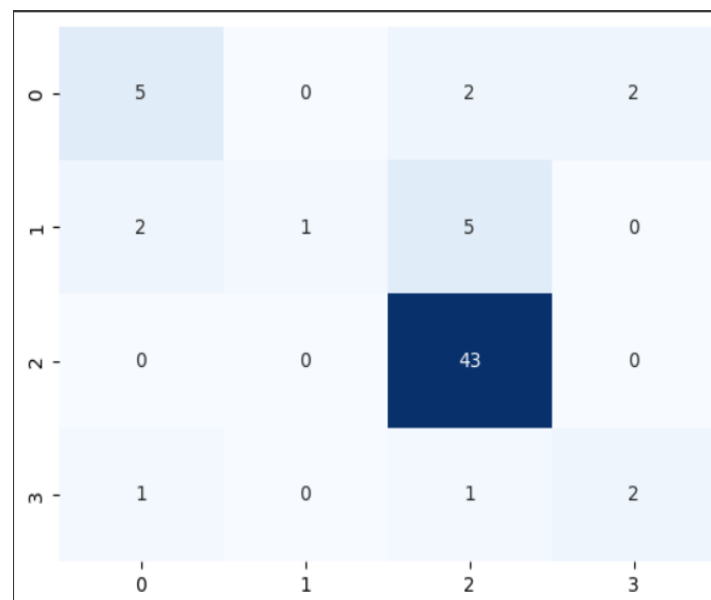


Fig 40. Confusion matrix for ResNet101.

## 8 Project Management

### 8.1 Project Schedule

The project schedule is used to ensure that the project is completed on time. Every week, we must review the project's schedule and ensure that it is followed. We have two deliverables: the approval of an ethical application on June 6<sup>th</sup>, 2023, and the submission of the final project on August 8<sup>th</sup>, 2023. The entire time allotted to this assignment since the beginning of the semester concludes on August 8<sup>th</sup>, 2023. We used a Gantt chart to indicate how we will complete the assignment over the following weeks.

Task 1: Data Collection

Task 2: Project proposal

Task 3: Ethics approval

Task 4: Literature review

Task 5: Code execution

Task 6: EDA and Pre-processing dataset

Task 7: Checking for the suitable machine learning models (Implemented classifiers, but not suitable for our dataset)

Task 8: Deployment of VGG-16, VGG-19, ResNet-50 and 101 model

Task 9: Results and conclusion (Finding the best model by its accuracy and loss)

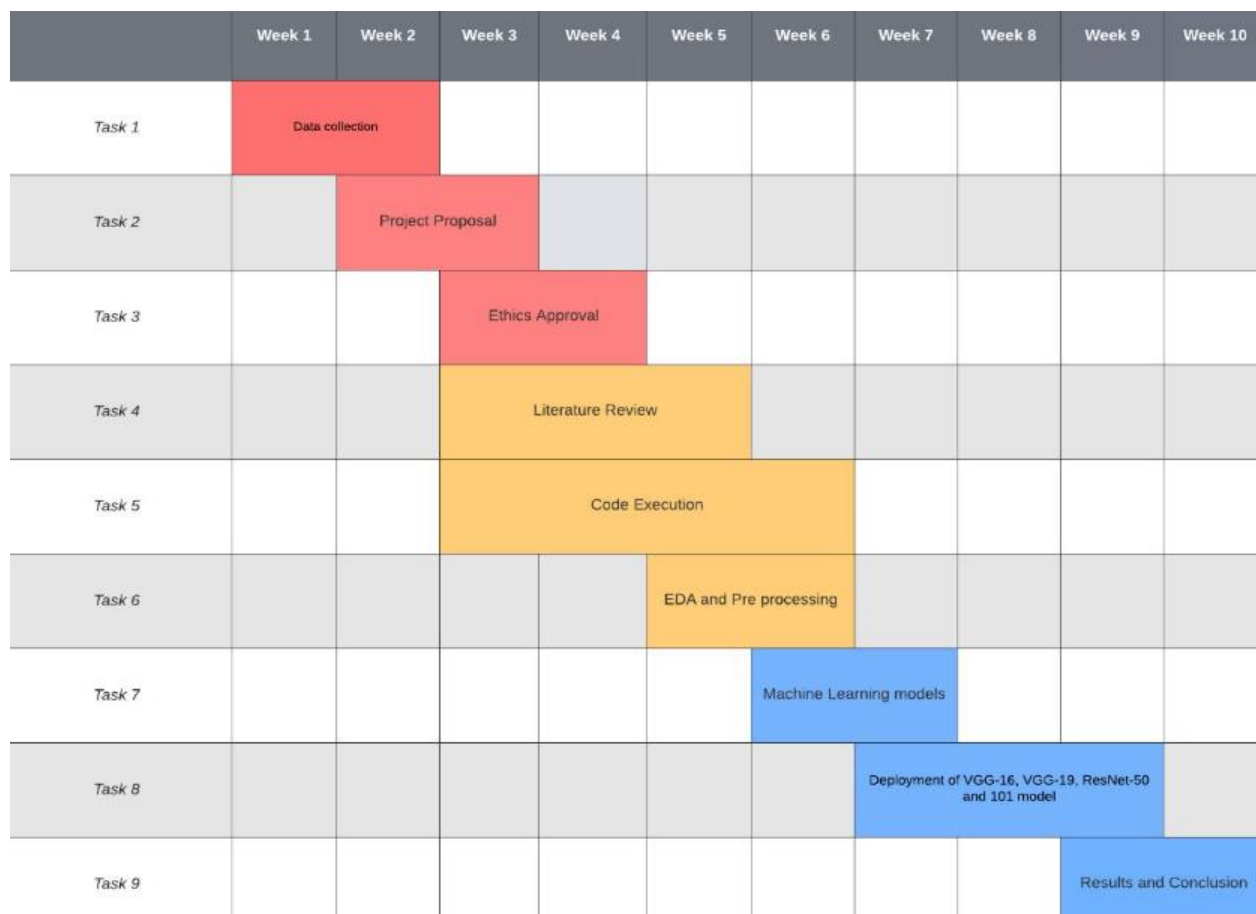


Fig 41. Gantt chart for all activities.

## 8.2 Risk Management

Apart from freely available dataset, the program never held any personal information about anyone. For this project, I obtained an image dataset. I had various problems putting the dataset into the platform. I had numerous problems while importing the enormous collection of image datasets. I had many issues while importing libraries using TensorFlow, and I also experienced many machine shutdowns and latency while running the Epoch to calculate the accuracy of the model. Also, there arises too many suspensions when creating visualization using tensor board. The only risk that developed during the project was time management; nevertheless, thanks to the supervisor's guidance and support, this issue was addressed and resolved effectively.

## 8.3 Quality Management

The majority of the side was created in Python. Python is an excellent option for tackling AI challenges because, after careful consideration of the process and methods, availability of the required additional information, and the presence of open-source libraries, has come to the conclusion that the Python programming language is the most suitable choice for implementing the discussed approaches and relevant technologies. The following critical step is to select acceptable libraries for such algorithms. TensorFlow and Keras are open-source tools developed by Google to handle AI and ML difficulties. The TensorFlow library is used to import and implement the primary algorithms ImageNet, ResNet-50, ResNet-101, VGG-16, and VGG-19. Given Google's notoriety, the libraries chosen are of the highest quality. In terms of platform and tools, the Python application is completely coded and then executed in Google Colab Pro. The Colab Pro has a large GPU which supports 16 GB. The essential steps have been highlighted with comments to help you understand the coding lines.

## 8.4 Social, Legal, Ethical and Professional Considerations

To submit the project proposal, it is mandatory to get ethics approved from the university. During this approval process the questionnaire is filled and submitted. Later this should be approved by the project supervisor and project head. Ethics form will be completely examined and approved by the supervisor and head of the project if there is no significant risk. The ethical criteria of the project align with the standards of the institution, thereby resulting in a direct outcome. Projects incorporating machine learning often give rise to social, legal, and ethical considerations due to the utilization of datasets that contain sensitive information and have the capacity to identify persons. The use of the dataset in this study which was derived from a freely available website, rendering it easily obtainable for scholarly examination and investigation. Consequently, no ethical or legal concerns have arisen in relation to its utilization. As a researcher, my main purpose is to conduct extensive and precise research in accordance with acknowledged scientific methods and to refer to ongoing and future development procedures. If I were a patient, I would work and meet with doctors to learn about the disease, drugs, and procedures that work. I continue my research after professional review.

Throughout this investigation, all pertinent sources have been appropriately referenced in the bibliography section, and adequate acknowledgment has been given whenever information derived from these publications.

## 9 Critical Appraisal

In general, implementing computer vision by using neural networks or deep learning architecture is a challenging task. This project's objective was to build an improved Deep Learning model that could determine the stage of cervical cancer based on pap smear images and do this analysis in real time.

Locating an appropriate dataset proved to be somewhat challenging. Applying a suitable pre-processing methods to it within the time restriction that was allotted for the project was a challenging endeavour in and of itself. Also, learning about the new topics like CNN, VGG, ResNet layers, transfer learning, Tensorflow, Keras in a short span of time was challenging but due to this research I was very happy that I have learned about many topics and also learnt the time management.

According to the findings, VGG-16 is the most effective model for the categorization of images, with an accuracy of 94% and 93%, respectively. As an outcome of these efforts, more was accomplished than was anticipated.



## 10 Conclusions

The results of our work offer an automated approach for performing tests of cervical cancer cells. In this study, we proposed the utilization of a deep learning model to effectively categorize Pap Smear images into four distinct groups, namely NILM, HSIL, LSIL, and SCC. This model would employ the VGG-16 architecture, which is widely regarded as the most effective among the utilized models like VGG-19, ResNet-50 and ResNet-101. The utilization of the vgg-16 architecture in the model facilitated a notable decrease in the necessary parameter list. In the context of multi-class classification, the training and validation average accuracies are 94 percent and 93 percent, respectively. The present methodology exhibits positive outcomes and supporting healthcare professionals in conducting outreach screenings, provided that it undergoes further refinement and evaluation through the inclusion of additional real-world pap smear images.

### 10.1 Achievements

A robust methodology for the identification of cervical cancer cells has been developed following extensive investigation and analysis of numerous academic literatures, alongside careful evaluation of patients' clinical presentations and symptoms. Highly efficient architectures such as VGG16, VGG19, ResNet50, and ResNet100 are employed for the purpose of cancer cell detection. In conducting this research, it is advantageous to have a large dataset. The findings of this study indicate that the models employed in this research have the potential to enhance accuracy and expedite processing, as well as facilitate classification, when applied to an image dataset. Furthermore, the utilization of deep learning techniques and transfer learning techniques is observed.

Through the process of evaluating these models, it becomes evident that they effectively eliminate financial barriers. Additionally, the implementation of an automated diagnosis system is advantageous in terms of time efficiency, as it speeds up the analysis of cancer stages and facilitates timely treatment.

### 10.2 Future Work

Establishing a link between LBC Pap smear pictures and the data from the accompanying biopsies is one way to improve the diagnostic accuracy and confidence offered by the model. In addition, the system has the ability to identify detailed abnormalities with better precision and improve its overall diagnostic skills if data that is produced from a variety of imaging modalities, such as colposcopy or fluorescence imaging, is incorporated into it. This would allow the system to take advantage of its full potential. It would be possible for medical professionals to have a better understanding of and ability to validate diagnostic results if the model's decision-making process was clarified more thoroughly.

Correlating LBC Pap smear images with matched biopsy data would improve the model's diagnosis precision and confidence. In addition, by merging data from many imaging modalities, such as colposcopy or fluorescence imaging, the system may be able to detect more subtle irregularities and perform more diagnostically as a whole. As a result

of more clarity on the model's decision-making process, healthcare practitioners would be better equipped to interpret and validate the model's diagnostic outcomes.

Finally, conducting extended research endeavours to observe and assess patient outcomes and monitor the impact of the model on patient care and treatment efficacy will provide invaluable knowledge for optimizing and maximizing the system's functionality in real-world clinical environments. The automated multi-class diagnosis system can develop into a complete and widely used tool, successfully enhancing cervical cancer screening and diagnosis for the benefit of women's healthcare globally, by taking into account these potential improvements and conducting additional research.

## 11 Student Reflections

In terms of scope and complexity, the execution of this project differed greatly from any previous experiences I had experienced. Nonetheless, it provided an excellent opportunity to broaden my learning and expertise, both of which will surely be useful and valuable in my future academic endeavours and profession. Furthermore, I claim that, despite some minor changes I had to make due to time constraints, I successfully handled the project with a high degree of organization. This method is based on my adherence to a planned management plan, as well as regular communication with my supervisor. This allowed me to tap into their depth of knowledge and receive weekly assistance.

In addition to the obvious findings, I offered insightful recommendations for future initiatives related to the subject. These recommendations were developed based on a detailed study and considerable research undertaken during the project. The inclusion of a section on future work in the paper aided in widening my understanding of other models. Another notable achievement is the progress made in the field of computer vision and the accompanying solutions that can be built. The TensorFlow object identification API gave me several ideas for prospective hobby projects.

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**Appendix A – Project Specification**

| Meeting | Topics Discussed                                                                                                                                                                    | Date       | Time    |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------|
| 1       | Supervisor recommended some topics to consider for research and asked to read academic literatures that is related to the research as a preliminary step to finish ethics approval. | 23-05-2023 | 10AM    |
| 2       | Explored the objectives related to the research and discussed on how to write literature review in report.                                                                          | 05-06-2023 | 12:30PM |
| 3       | Advice is given on how to write the complete report and code.                                                                                                                       | 04-07-2023 | 11AM    |
| 4       | Meeting with Anuja Talekar where she suggested me on how to proceed with report writing.                                                                                            | 01-08-2023 | 3PM     |
| 5       | Made correction in the report part and suggested few changes in code as well.                                                                                                       | 02-08-2023 | 10:30AM |

## Appendix B – Ethics approval certificate:

Classification and identification of cervical cancer using CNN methods.

P158757



### Certificate of Ethical Approval

Applicant: Thanmaya Badhepudi  
Project Title: Classification and identification of cervical cancer using CNN methods.

This is to certify that the above named applicant has completed the Coventry University Ethical Approval process and their project has been confirmed and approved as Medium Risk

Date of approval: 06 Jun 2023  
Project Reference Number: P158757



## **Appendix C –Dataset and code:**

### **Dataset:**

<https://data.mendeley.com/datasets/zddtpgzv63/4>

### **Code:**

[https://drive.google.com/drive/folders/1\\_kmUsj38D1gnljVjWN1s1BXz8795lIT3?usp=drive\\_link](https://drive.google.com/drive/folders/1_kmUsj38D1gnljVjWN1s1BXz8795lIT3?usp=drive_link)